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A THESIS REPORT

ON

**Advancing Generative Adversarial Network
Architectures and Cross-Modal Learning Strategies
for High-Fidelity Text-to-Image Synthesis**

PREPARED

BY

Md Saidain Islam Polin (Roll No: 1908036)

Enamul Hasan Shagato (Roll No: 1908058)

DEPARTMENT OF MECHATRONICS ENGINEERING

JUNE 2025

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Department of Mechatronics Engineering



A THESIS REPORT

ON

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Advancing Generative Adversarial Network Architectures and Cross-Modal Learning Strategies for High-Fidelity Text-to-Image Synthesis

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A thesis report is submitted to the Department of Mechatronics Engineering, Rajshahi University of Engineering & Technology in partial fulfillment of the requirements for the degree of Bachelor of Science in Mechatronics Engineering.

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Department of Mechatronics Engineering



CERTIFICATE

This is to certify that the thesis work “**Advancing Generative Adversarial Network Architectures and Cross-Modal Learning Strategies for High-Fidelity Text-to-Image Synthesis.**” by **Md Saidain Islam Polin**, bearing Roll no. **1908036** & **Enamul Hasan Shagato**, bearing Roll No. **1908058** has been carried out under my supervision in partial fulfillment of the requirements for the degree of Bachelor of Science in Mechatronics Engineering.

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Declaration

We, **Md Saidain Islam Polin** and **Enamul Hasan Shagato**, affirm with full responsibility that the thesis entitled “**Advancing Generative Adversarial Network Architectures and Cross-Modal Learning Strategies for High-Fidelity Text-to-Image Synthesis**” is solely authored by us. This thesis is presented to the esteemed Department of Mechatronics Engineering, Rajshahi University of Engineering & Technology, as part of our journey towards obtaining the Bachelor of Science degree in Mechatronics Engineering. We gratefully acknowledge the guidance provided to us by **Md. Faisal Rahman Badal, Assistant Professor** in the Department of Mechatronics Engineering, Rajshahi University of Engineering & Technology.

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May the Almighty give us with every blessing in this world.

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Abstract

This thesis is an enhancement to the model that is known as Attentional GAN in the high quality text to image synthesis. The primary aim of the current paper is to improve alternative images that are produced on the basis of text descriptions through the prism of quality, diversity, and semantics fidelity. The above-mentioned gains, starting with the multi-scale attention mechanism, progression learning, all the way to the multi-critic model, all contribute to the fact that the model is in a position to generate more photorealistic and diversified outcomes. We understand what is the review of existing technologies of T2I and then use, train CUB_200_2011 bird data, and enhance AttnGAN model. It was experimentally tested with the aid of multiple performance measures, including IS, FID, and perceptual loss, visual analysis, and diversity of generated images. There is reported big improvement of the generated image quality of the generator with increasing training of the model (IS study increased by 0.45 to 3.9), which implies that the model is generating more diverse and more realistic images. In addition to that, the multi-critic model provides the output image to be both realistic and consistent across different resolutions. Even though the suggested model displayed otherworldly outcomes, the detailing at the finer level and the cost of training computation were difficult. The work will be in improving to such limitation, however, it will equally be in using other resourceful training, and which will further extend the model to more complex and wide areas.

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List of Abbreviations & Symbols

GAN	Generative Adversarial Network
VAE	Variational Autoencoder
T2I	Text to Image
CV	Computer Vision
NLP	Natural Language Processing
AttnGAN	Attentional GAN
AI	Artificial Intelligence
FID	Frechet Inception Distance
IS	Inception Score
VGG19	Visual Geometry Group 19
RNN	Recurrent Neural Network
CLIP	Contrastive Language-Image Pretraining
CGAN	Conditional GAN
DCGAN	Deep Convolutional GAN
PGAN	Progressive GAN
GPU	Graphics Processing Unit

Chapter 01

Introduction

Text-to-image generation is an active area of AI research over the past few years. The combination of NLP and CV evokes a lot of prospects in the field of content generation: visual storytelling, design automation, custom media, and so forth. Another common one in this environment is the GAN, which has altered our way of thinking in regards to image synthesis. GANs rely on a two-network: the generator and the discriminator that runs an assessment. The need of this adversarial training in creating realistic images arises because of the fact that the output of the generator is iteratively trimmed.

Among the most important distinctions of the T2I synthesis of the GANs, there is AttnGAN. AttnGAN uses attention mechanism to constrain the generator to focus on various sub-regions of the prompt during its image generation. However, irrespective of the latter, AttnGAN continues to falter in producing images with an abundance of details.

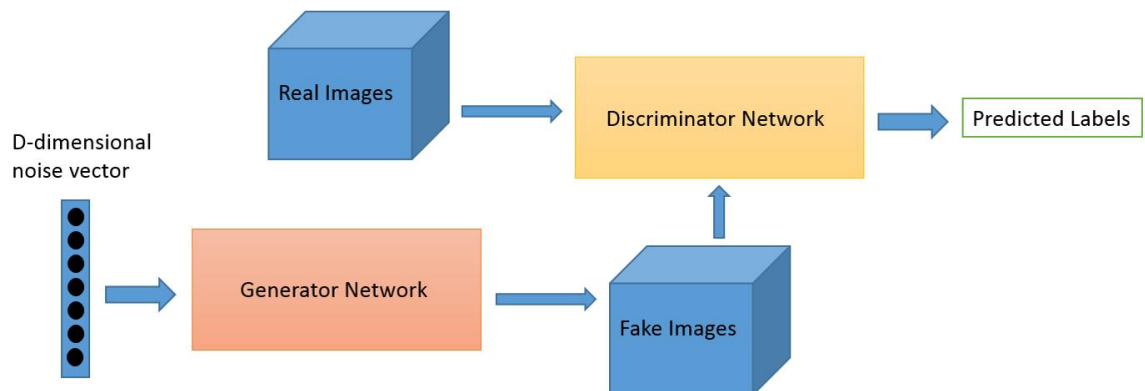


Fig 1.1: Working process of a basic GAN model.

In this work, we extend the AttnGAN model by introducing a number of architectural innovations. These techniques shed light on how to customize detail of the resultant photos through utilizing multi-scale attention mechanism, progressive training policy, multi-critic and so on. This thesis will contribute with human perception based evaluation of the generated content instead of pixel-wise perceptual loss. Finally, the target of this work is to design an architecture that can synthesize high-definition, fine-grained photos from detailed natural-language descriptions and contribute to the emerging area of T2I synthesis.

1.1 Overview

The performance of T2I has become an interesting issue in AI research field, especially for the CV. The field was revolutionized in 2014 when Ian Goodfellow and his collaborators introduced GANs which had breakthroughs in development by presenting a framework that worked in terms of two neural nets[23]. The discriminator distinguishes generated data from real data, whereas the generator produces realistic looking data. An important recent application of GANs in T2I synthesis is AttnGAN which leverages attention mechanism to steer the generation process. AttnGAN can attend to certain thematic components and progressively refine them during the generation process. However, despite the advancements made by AttnGAN, there are still challenges regarding image quality, particularly in maintaining a balance between fine-grained details and global structures, as well as handling the complexity of text descriptions in high-resolution images. This thesis aims to enhance the AttnGAN model by integrating four key advancements: a multi-scale attention mechanism, a progressive training approach for the generator, a multi-critic model, and the incorporation of perceptual loss.

1.2 Background and Motivation

The ability to generate realistic images from a textual content is widely appealing in diverse applications such as entertainment, virtual reality, e-commerce and accessibility. There is such a great demand for generating high quality images on-the-fly in these applications, such as online shopping, virtual product demonstration, or creative creation. Existing methods of synthesizing images from natural language descriptions have been dominated by simple models which produce low-fidelity images. Although models like GANs show some promise, T2I synthesis is still a challenging task, one that involves various complexities in understanding and translating language to visual.

AttnGAN had also a significant breakthrough in this direction where they utilized the attention mechanism at the generation side of images. Although the quality of the generated images of this method is much better than that of previous methods, it is still difficult to generate images that not only preserve lots of details (e.g., textures) but also preserve the high-level structure (e.g., shapes and arrangements) in a consistent way. And

that the generator outputs images and as it is in general using one single discriminator, the output might not be as refined since the feedback quality can be worse for the training.

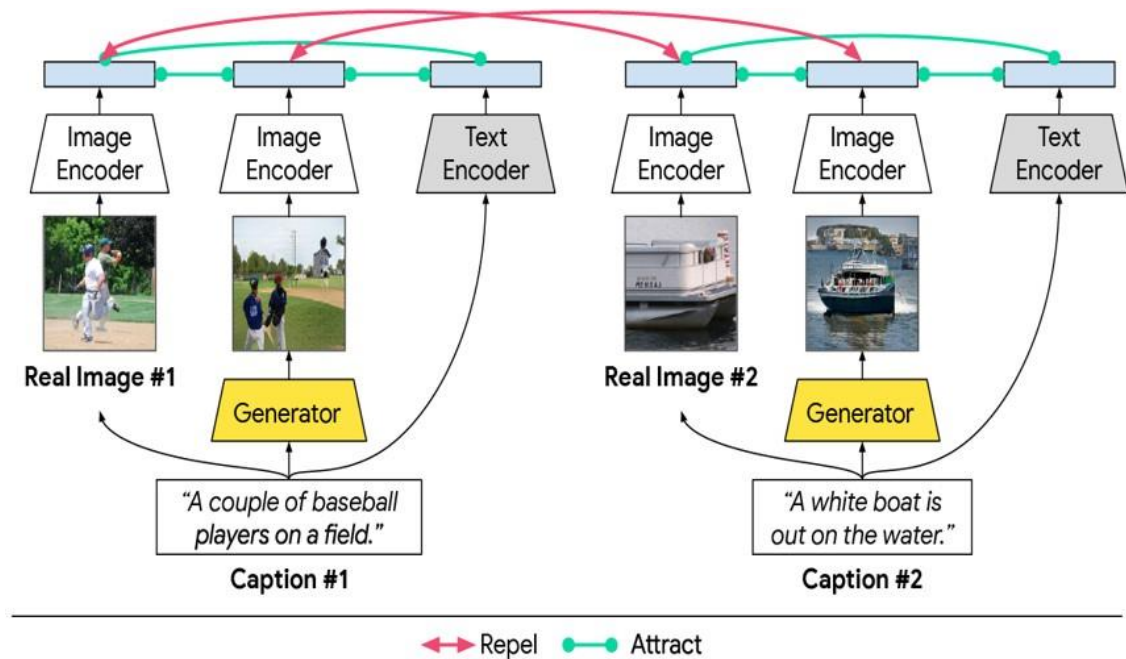


Fig 1.2: Basic GAN Architecture Workflow[57]

The reason for working on this topic comes from the observation that current algorithms such as AttnGAN still struggle when dealing with images that are complex and/or high-resolution. With the proposed multi-scale attention mechanisms, we are enabled to emphasis fine-grained details as well as larger structures across different resolutions. In addition, the use of progressive training will enable the model to produce images in a progressive manner, and intermediate results will get more precise at each stage. The addition of multiple discriminators to the multicritic model can lend more specific feedback, thus enabling a more nuanced assessment of the generated image. Last, perceptual loss can be adopted to serve as a surrogate for the pixel-wise loss function for better calibration of the synthesized image to human perception in terms of structural and texture contents rather pixel-based score matching.

1.3 Objectives

This thesis's principal goal is to further the AttnGAN model by introducing a series of architectural and methodological improvements. The key objectives follows:

- i. Design novel GAN architectures by generalizing AttnGAN for high resolution, semantically coherent text-to-image synthesis.
- ii. Mitigate the computational burden of diffusion models by exploiting GANs for the fast and straightforward high fidelity image generation.
- iii. Develop methods to seamlessly transfer information between text and image modalities, ensuring the accurate alignment of text descriptions and visual content.

1.4 Research Methodology

This study's research technique was used to further the AttnGAN model for high-fidelity text-to-image synthesis involves several key steps, each critical to improving the accuracy and quality of generated images. These steps are outlined below:

1. **Literature Review:** The first step of this study is an extensive literature survey, that surveys the recent progresses in T2I, especially the architecture of AttnGAN. This includes current state of the art GAN-based models, attention models, progressive training methods, multi-critic models, and perceptual loss functions. Investigation provides a basic description of the techniques, shows deficiencies in the current model and points out the future work of the AttnGAN model.
2. **Data Collection:** An important stage of the study includes the generation of data suitable for training adapted AttnGAN model. The datasets are usually composed by image-caption pairs, giving rise to a large variety of textual descriptions and image examples. Datasets such as Microsoft COCO and Oxford 102 Flower datasets will be utilized to train and test the model. It is a perfect dataset for testing text-to-image generation models as there are detailed textual descriptions provided in addition to images.
3. **Data Preprocessing:** The raw dataset that is taken as input, is processed to ensure data cleanliness and consistency for manipulating the model. This stage involves cleaning (e.g., removing duplicates, reconciling inconsistencies in the data) the text and images and structuring their text and image formats in a consistent manner. Also all input captions are the text is normalized, i.e, converted to lower case and certain punctuations removed.
4. **Feature Engineering:** In this step, information of interest is generated from

image and textual descriptions. For the text, important aspects such like object names, features, and geometric relations are detected as a meaningful input for the attention mechanism. Visual information encoding is captured from the images such as the textures, shape of the objects, and structure patterns. These features will serve as input to an improved AttnGAN model such that the generator is trained to produce images that would correspond to the input text descriptions.

5. **Algorithm Development:** The gist is that we extended and implemented AttnGAN. These involve the use of multi-scale attention mechanisms, progressively training the generator and the use of a multi-critic model. Observing that the multi-scale attention mechanism can focus on both small-scale and large-scale details of generated images. The progressive training methodology allows the generator to primarily learn the generation of low-resolution images and then gradually refine them to higher resolutions. The multi-critic model uses a set of discriminators, each assessing a different dimension of the generated image (texture, structure and overall visual quality).
6. **Model Training:** Then we train the enhanced AttnGAN model (utilizing the multi-scale attention, progressive generation and multi-critic feedback), with the preprocessed dataset. The model is learned using off-the-shelf DL frameworks, and concentrating on the network's capability for enhancement. Furthermore, we carry out a perceptual study to see how realistic the synthesized images are, and whether they are human-expected or not [33].
7. **Deployment and Monitoring:** After the successful user testing and adjustment, the model is sent out for wider use. The performance of our method is continually verified to guarantee the robustness of it to generate high-quality images from textual descriptions. This continual supervision is crucial to detect any potential problems in various real-life cases, and to guarantee the model's robustness.

1.5 Project Planning

A good project planning is therefore very important for the successful implementation of this study. The design makes the research to be conducted in a systematic and orderly way and with clear milestones, task and time lines. The project plan in this study is a continuous batch framework that spawns out the smooth flow from a literature review and data collection into model development, model training, model evaluation, and final model deployment. The process outlined above will be performed in an iterative manner and periodic review will be made of the model enhancement progress. The schedule reports several work stages

(preliminary study, model development, test) and for each of them, the resources and times needed to reach the expected results.

1.5.1 Work Plan

The workplan of this research is presented in terms of different phases to accomplish a specific objective of the project. The first stage of the work plan is dedicated to consist a comprehensive literature review that provides an insight into GANs. This will correspond to approximately the 1st month, for reading and exploring existing papers, models, and techniques in order to lay a foundation for later work. After the literature review, the next stage is data gathering and data preprocessing. This stage will involve the extraction, cleansing, and preparation of the appropriate datasets for model training. This process should take about two months, as we'll be looking only at high quality, well-labeled data.

The model development phase will follow on and is expected to last in the region of three months. This stage will involve the development and training of the improved AttnGAN model with multi-scale attention, progressive training, and multi-critic model. This time, I'll fix the perceptual loss as well. Every part of the model will be build tested and refined hole and each and by each to make it work just the way it is meant to. Two months Next 2 months Training / Evaluation With the model trained, training and evaluation will be the main focus for the next two months. At this stage, we train the model on pre-processed data, and evaluate its performance using numerous testing metrics. The quality of generated images will also be evaluated qualitatively by obtaining the opinion on the images by users.

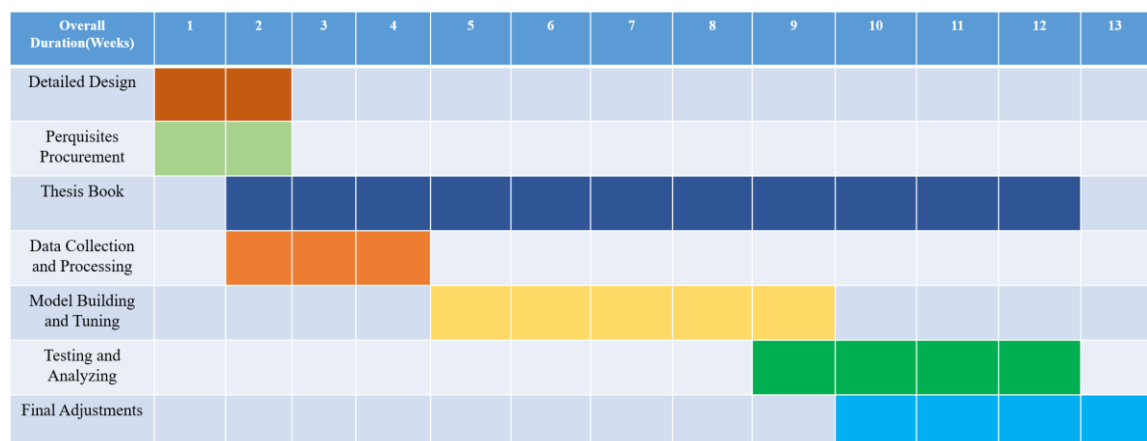


Fig 1.3: Gantt Chart

The last phase of the work plan is user testing and rollout. The model will be revised based on user feedback to improve the model performance. This phase will last one month, at which time the model will be implemented for wider use. Continual maintenance will be performed to help ensure that the model sustains high performance when used for real-world applications.

1.6 Organization of the Report

Here is a breakdown of the report:

1 Introduction The first chapter will introduce the topic of the study: T2I, and the problems which are resolved in this work First Chapter. It also states the objectives, methodology and report's organization.

The body of literature on GANs and T2I is reviewed in Chapter 2, with focus on current models including AttnGAN and the need for future research. This chapter also gives a brief overview progress in attention mechanisms, progressive training and perceptual loss.

The method used in this work, which focuses on enhancing the AttnGAN model, is discussed in Chapter 3. It comprises the multi-scale attention model, the progressive training of the generator, the multi-critic model, and the perceptual loss.

The experiments to evaluate the effectiveness of the proposed modifications are presented in Chapter 4. It contains the data employed, the evaluation method, and the results obtained quantitatively and through a visual assessment. This section demonstrates that the proposed multi-scale attention and multi-critic feedback mechanisms are effective.

Chapter 5 concludes with the simulation results and comparisons between the baseline model and the proposed model. Then the influence of the methods on the photo generation course is analyzed, on two aspects: fine details generation and overall image structure generation.

6 discusses the limitations of the study (in the areas of stability of the model, computational cost, and integration of the perceptual loss function). It also indicates possible avenues to address these shortcomings in future research.

In Chapter 7, possible future development for improving the AttnGAN model is explored, alongside with proposed places for improvement and new methods, which could potentially improve T2I quality. This section also highlights the findings and their implications in a larger context and future potential for the study in other disciplines.

Chapter 8 concludes the research, summarizing the main outcomes and discussing the contribution to the text-to-image synthesis domain. It also describes the direction for future research and the possible avenues to take to further enhance the capabilities of generative models.

1.7 Summary

The focus of the research is on extending the AttnGAN model towards high-fidelity text-to-image synthesis, and solving the critical problems in generating realistic and detailed images from text descriptions. This chapter introduced the research by discussing the motivation for our research, including the drawbacks of the state-of-the-art models and desired improvements, i.e., multi-scale attention, progressive training, a multi-critic model, and perceptual loss.

The research design was also explained along with an introduction of the design, training and test. The chapter ended with a description of how the report is organized – that is, the structure of the following chapters which in themselves, regarding the literature review and the methodology, the results and finally the conclusions, focus on themes from the research.

Chapter 2

Literature Review

The related work section starts with a review on GANs and its developments, then moves on to existing methods used specifically in steering text-to-image synthesis. This review also extends to the state of the art in attention, on-the-fly training, and perceptual loss all of which are core in this thesis. This chapter thus establishes a critical framework through which the present research will be located, providing the background and context of AttnGAN, along with a cut-out of existing approaches, and paving the way for our proposed method of AttnGAN extensions.

2.1 Introduction

The fertile field of generative models, especially GANs, has transformed how we think about generating new images, and nowhere is this truer than in the notoriously hard task of generating images from words. Goodfellow introduced GANs. Although this vanilla GAN architecture has been fundamental, generating images from textual descriptions demands more complexity to bridge the communication gap between the visual appearance and the spoken description.

There are many variants and extensions of the original GAN that afford improvements. Among them, the cGAN has been one of the most important advancement to enable the generation of the images from specific inputs. This conditioning could be as simple as a label, a "cat" or "dog", or something more complex, a text description [1]. The development of CGANs has opened the possibility of generating images that possess certain conditions, and it has become one of mainlines for T2I.

As there was increasing demand for generating more diverse and high quality images, DCGANs became the ideal solution for generating images. The fully connected layers were replaced by DCGANs. The generator of this model should be resized from 64x64 to 128x128 by taking the 64x64 images and doubling their size. This is one of the better GANs that are able to extend DCGANs to larger images and therefore sooth the text-to-image task and whatever realted. But despite DCGANs being some kind of revolution for image generation, they were still very far away from photo creation with

the natural language description to the desired photo. In light of this, AttnGAN [43] was proposed, which can be addressed by the attention.

StackGAN [3] afterwards proposed the two-way generation manner after the AttnGAN. The first stage of producing the low-res images, and staked_2 refined them up again. This progressive method could help the model handle complicated details much better, and therefore produce more realistic images. StackGAN paved the way for models based on more than one stage or layer of generation, improving images at each step of the pipeline.

CycleGAN as introduction met the requirement that of image to another. Though originally aiming at applications such as photo manipulation and style transfer, CycleGAN has also had an impact on text-to-image generation models by encouraging the use of cycle-consistent losses for image generation. This idea guarantees the reversibility of domain transformations.

On the quest for generating better designed and high resolution images, PGANs came into picture with their unique approach of gradually increasing resolution of images while training. PGAN start with low-resolution images and then build up where the training is stable and image quality high and detailed. This technique is successful in synthesizing high-resolutions images and has influenced other text-to-image models to generate fine-grained photo-realistic images.

Lastly, BigGAN [2], one of the latest GAN architectures, makes use of large-scale models and strong computational resource to produce realistic and high-resolution images. BigGAN does break new ground in terms of image quality and diversity, through the use of larger models and batch sizes, but is also particularly computationally expensive. Although BigGAN is not a T2I model, it has inspired subsequent works for generating more complex and more realistic images.

2.2 Literature Review

The text-to-image synthesis task has gained a considerable attention in recent years, particularly after the success of GANs. Creating realistic pictures given text descriptions presents a particular challenge since it requires both understanding the semantics of the text and translating that understanding into coherent, high-quality visual depictions. As

the GAN-based image generating becomes more and more mature, many models have been brought forward to tackle the special challenges presented by the task of text-to-picture synthesis. The literature review studies GAN-based methods with special focus on those applied for text-to-image problems [36]. cGANs in which the discriminator and generator is conditioned by additional information formed the basis of the early work in text-to-image synthesis [46]. This becomes an essential structure for text-based picture generation as it allows the representations to produce images based on specific attributes, e.g. labels or other forms of structured conditioning. Despite the use of cGANs as a powerful base, they still struggled to produce images that could truly match the complexity of the textual input. This change meant this form of GAN was better able to produce images of higher quality and better structure and led to being DCGANs and ideal candidate for image generation tasks. DCGANs were important in enhancing image realism but did not tackle the core issue of matching text and images.

Later, the field took a further leap with AttnGAN, where a key to this success was the addition of attention based mechanism. Fig. 1 (b): The attention model allows the model to produce a more detailed and contextually consistent image by focusing on different parts of the text at different generation step. It also uses a new developing approach where image generation happens in multiple stages, getting bigger resolution and more real at each successive stage[43].

Building on the results from AttnGAN, StackGAN proposed a two-stage process for image synthesis. This gradual refinement makes the overall image quality better by first covering up the big structures and then carving in the small ones. StackGAN proved to be useful to stabilize the training and to generate high-quality images, especially when conditioning on more complex descriptions [45]. Such a two-stage generation process efficiently addresses the problem of synthesizing images based on complex text descriptions.

CycleGAN opened up a new room in image synthesis due to the cycle-consistency loss. Although CycleGAN itself mainly concentrated on image translation, the concepts of cycle-consistency and image transformation have inspired text-to-image models to be able to do bi-domain translations consistently. This cycle-consistency loss terms tends to force structural details to reintroduce themselves, increasing the content consistency for generated images.[44]

Another major contribution was introduced by PGANs with a technique of progressively growing the image resolution during training. PGANs initially create the low-resolution images, which are then gradually refined as the resolution converges [42].

Hybrid models building has also become popular in recent years with the born of such structure involves GAN may deals with other networks like VAE and RNN[3]. For instance, VAEs are useful for generating diverse images, while RNNs are useful to learn about the sequentiality of text [43].

Along with these GAN-based developments, the use of perceptual loss has improved the quality of images. Classical pixel-wise loss functions were not good at producing images that look pleasing to human eyes.

However, for T2I the (challenging) story is different. The correspondence between textual descriptions and visual contents are far from perfect and it is challenging to generate fine-grained images while keeping them in consistent with the text. Furthermore, computational cost and stability of training in the high-resolution image generation are still challenging.

Table 2.1: Related Works and Their Literature Review

Ref.	Published Year	Methodology	Findings	Limitations
[51]	2022	CMAFGAN enhances face synthesis with detailed semantics from attribute-word sequences.	CMAFGAN framework uses attribute-word sequences for face synthesis, offering richer semantic information than attribute vectors.	The reliance on attribute words may introduce challenges if the attribute descriptions are incomplete or ambiguous.
[52]	2023	Incorporating CLIP, ALIGN, and FILIP to align vision and language embeddings.	Cross-modal pre-trained models such as CLIP, ALIGN, and FILIP excel in aligning vision and language embeddings	The efficacy of unified training in DiffDis may require extensive computational

			for various downstream tasks.	resources and longer training times compared to single-task models.
[53]	2022	Tackling GAN Flaws to addressing architecture, supervision, and fusion issues.	Existing text-to-image Generative Adversarial Networks (GANs) often face three key flaws related to architecture entanglements	The generalization of DF-GAN to diverse datasets and textual descriptions needs further evaluation to assess its robustness across different domains.
[54]	2021	The GAN architecture incorporates both local and global language representations through the CGL-GAN Framework.	The study suggests GANs based on Cross-Modal Global and Local Linguistic Representations.	Higher-resolution picture generation using GAN-based models frequently leads in illogical outcomes and unreliable training.
[55]	2023	Models based on GANs frequently display inconsistent training and provide absurd	Slow synthesis processes, the requirement for a lot of training data and parameters, and the complexity of	GALIP's performance relies heavily on the pretrained CLIP model, which may limit

		outcomes when producing higher-resolution pictures.	managing synthesized visual elements make it difficult to generate high-fidelity complex pictures from text.	its applicability in scenarios where CLIP is not suitable.
[56]	2021	Developing a discriminator capable of distinguishing between content and style components.	Existing T2I synthesis models have trouble producing the text-related content portion of the picture while simultaneously adjusting the resulting image's appearance.	The generalization of the proposed modality disentangled discriminator to diverse datasets and text descriptions should be further evaluated to assess its robustness.

2.3 Recent Methodologies

Recent years have witnessed an explosion in research of **text-to-image synthesis methods**, particularly **based on GANs**. Generating visuals from text cues, however, remains challenging, as it requires an understanding of the content of the text as well as the ability to represent it in an appropriate sensible visual form. In order to combat these challenges, several new procedures have been proposed by introducing novel ways and structures that enhance text-image alignment, enhance picture quality and increase the set of generated images.

One of the major progresses of current methods is the construction of multi-scale attention mechanisms. Owing to this limitations of non-attention-based approaches, recently attention mechanism has been widely used for many NLP tasks which involve interactions between input words and many output generation processes, e.g., translation

13 or machine comprehension. A recent development multi-scale attention allows the model to simultaneously attend at multiple image scales. At lower levels, the model identifies finegrained details (e.g. textures or patterns) and at higher levels it attends to coarser features). In this way, we ensure that both small details and large-scale layout are preserved in the resulting image, and hence produce more consistent and realistic outputs [39]. Models such as AttnGAN have adopted such multi-scale attention process, in which the generator can attend to different parts of the text description at different levels of image generation. This attention mechanism makes it possible to have a more direct mapping of text and the visual content, resulting in higher fidelity images.

11 However, this is not an ideal or full proof way for measuring different components of the image, such as texture, structure and visual coherence. For that, more recent approaches have presented multi-critic models in which multiple discriminators are applied to check different features of the image. These discriminators specialize in aspects of the image like texture, spatial receptive field, or global visual quality [47]. A second major methodological development has been the design of perceptual loss functions which have since that time and going forward been a key for sharp and coherent image generation. Although pixel-wise loss have been effective in some situation, it cannot characterize perceptual properties of images such as texture, style and structural coherence [38]. The loss function compares the structural and semantic differences between images, as opposed to their pixel differences, thus being more consistent with human perception. Perceptual loss has consequently driven them to both visually realistic and human observer friendly as well. The transition from pixel-wise to perceptual loss has proven particularly important in tasks like text-to-image synthesis, wherein the precise pixel values do not matter but it is contextually and perceptually realistic.

52 The field has gone a step further with the development of hybrid models that combine GANs with other neural network architectures like VAEs. VAEs, since they enable the model to capture a wider range of visual variations, are quite useful to generate diverse outputs [48]. Taking advantage of the merits, VAEs and GANs have been merged by researchers: VAEs can generate diverse pictures while GANs can generate high-quality images. Especially in text-to-image generation, this hybrid approach has shown some remarkable promise in increasing the freedom and reality of the newly generated images.

While such approaches advance, their models can ultimately synthesize high-quality, diverse, and contextually plausible images based on text inputs. The incorporation of attention mechanisms, progressive training, multi-critic models and perceptual loss has played a great effect on the effectiveness of text-to-image models. In addition, cross-modal learning, and hybrid models provide opportunities to further improve the performance and efficiencies of such systems. Despite this success, there is still a gap in training the models to generate more realistic and diverse images, especially when the textual descriptions are complex and ambiguous.

Table 2.2: Different Architectures and Methodologies

Methodology	Description	Impact	Strengths	Challenges
AttnGAN	Utilizes a multi-scale attention mechanism to generate detailed images from text descriptions.	Allows the model to create images that are more contextually accurate.	Allows fine-grained control over image features focusing on specific segments of the text.	Requires balancing multiple attention layers, making the model computationally expensive and challenging to optimize.
StackGAN	Generates images in two stages.	Generating more coherent and high-quality images by refining the low-resolution outputs.	Progressively refines images.	Training requires significant computational resources and can be difficult to fine-tune for different image resolutions.
CycleGAN	Focuses on image-to-image translation tasks, utilizing cycle-consistency loss.	Improves the consistency of generated images.	Ensures the transformation of images between different domains.	The cycle-consistency loss may sometimes reduce image fidelity, especially when transforming between complex domains like text and images.
VAE-GAN	Combines VAEs and GANs to improve both the diversity and quality.	VAE-GAN increases the variety of pictures produced.	Combines the strengths of VAEs and GANs to generate diverse and realistic images.	The VAE-GAN hybrid approach is computationally intensive, and the trade-off between diversity and image quality can be difficult to optimize.

2.4 Detailed of Reviewed Methodologies

AttnGAN's hierarchical attention mechanism focuses on different areas of the input text at different stages of picture generation by employing different attention layers. This is necessary in order to ensure that the generated image can resemble as closely as possible the unique properties described in the text. For example, when getting produced, the model may look at specific objects or attributes which are referred to in the description [4]. With the attention mechanism, the model can produce detailed and high-quality visuals for better understanding of the context of the text description. But even though AttnGAN's attention mechanism works well in most scenarios, it can still fail in the cases where the text is ambiguous or where there are intricate relations between objects and thus needs to be further polished.

In StackGAN the first stage synthesizes the low-resolution image by focusing on rendering the basic shape of the scene described in text whereas the second stage adds more content and refines the final image to high quality level. StackGAN can generate large pattern and fine detail of pictures since this progressive generation [49]. A big advantage of StackGAN is that it makes it easy to enable the generation of captions at different levels of abstraction; one major relatively simple change unlocks many different kinds of text-to-image generation tasks. However, this two-step strategy may result in problems of fine image detail refinement, especially the text input is ambiguous or without sufficient description. In addition, the model is computationally expensive to train especially if it is scaled up for high-resolution image synthesis.

CycleGAN, built for image-to-image translation, has inspired text-to-image models as well, with its innovative cycle-consistency loss. Such a loss function guarantees the "reversibility" of one domain to another transformation, which can preserve the consistency and coherence of the generated images. In the case of text-to-image generation, the purpose of this cycle-consistency loss is to force the model output images to not just resemble the text description but to also maintain highly salient structural details that can be transformed back into the input text. Although CycleGAN has exhibited promising results in image-to-image translation tasks (e.g., photo enhancement and style transfer), adapting this idea to text-to-image synthesis is non-trivial due to the difficult to preserving the fine-grained correspondence between the input text and images.

[5] In addition, CycleGAN's image-to-image translation may constrain its expressiveness in synthesizing diverse images from higher-level or more complex textual descriptions.

To address the limitations of traditional GAN models, VAE-GAN takes the advantages of VAEs and GANs together. Whereas GANs are very good for generating high-quality pictures, VAEs can generate a lot of variety in output because they learn a probabilistic distribution over the input. VAE-GANs Bringing these two architectures together, VAE-GANs are capable of generating realistic and diverse pictures, a quality that can be extremely valuable for text-to-image synthesis tasks that require a large amount of fidelity and variety [40]. One key advantage of VAE-GAN is that it can learn the underlying picture distribution, to allow a broader range of outputs. On the other hand, it can be hard to find a good balance between the number of produced images and their quality. Although being computationally heavy, VAE-GAN models have to be fine-tuned to obtain a perfect trade-off between realism and diversity.

State-of-the-art text-to-image models based on deep learning have recently integrated transformer-based architectures as well as GANs. Transformers, which were initially introduced for NLP tasks, have found successes on image generation tasks, e.g., text-to-image synthesis. These models utilize self-attention mechanisms to handle long-range dependencies between text input and generated images which help them to capture more complex text-image relationships. The amalgamation of transformers and GANs yields substantial models that are able to cope both complexity and diversity of textual descriptions. [9] Nevertheless, transformer-based models often have computational requirement that are not negligible, especially when we need to consider large-scale datasets and high-resolution image generation tasks.

The analyzed methods represent the evolution of GAN-based models for text-to-image generation, each one giving a different contribution to the field. StackGAN's two-stage approach enhances visual realism, AttnGAN's attention mechanisms model contextual relationships, and CycleGAN's cycle-consistency ensures the consistency of transformations. Text-to-image synthesis in the form of the transformers-based deep learning models is now promising even more complicated form of text-to-image synthesis, and VAE-GAN's hybrid approach between transformers and shallow models

generates richer and higher quality results. Yet, there are still concerns on the ambiguity re-surfing from textual descriptions as well as improving text-to-image attribute alignment and maximizing computational efficiency when scaling to high-resolution images.

2.5 Summary

In this chapter the state-of-the-art in text-to-image synthesis was systematically discussed, with a focus on recent trends including and going beyond GANs. The reviewed literature highlighted the remarkable contribution that various GAN-based systems have made to the problem of generating realistic pictures from textual descriptions. Models such as AttnGAN, StackGAN, CycleGAN, VAE-GAN as well as hybrids such as combining transformers with GANs were used to tackle the challenges in text-to-image generation. The multi-scale attention mechanism in AttnGAN helped the model focus on specific segments of text at various stages of picture composition, thus promoting better alignment between text and image synthesis.

Chapter 3

Design of Model Architecture

A very strong model architecture is required to achieve high-fidelity T2I, and this chapter outlines the design method to improve the AttnGAN model. Despite numerous advanced methods being employed, the overarching objective is to improve the model's ability to generate plausible, complex images based on textual descriptions. Specifically, the methods include a multi-critic model, progressive training, and a multi-scale attention mechanism. The primary aim of these improvements is to rectify the existing issues of the original AttnGAN model, which failed to strike the balance between fine-grained image characteristics and general image structure. In contrast, this new architecture seeks to enhance the generator to ensure it represents both low-level elements like textures and fine scale features and high-level concepts such as object relationships and scene compositions adequately. Furthermore, the adoption of multiple discriminators in a multi-critic framework is expected to yield more detailed feedback, enabling the generator to concentrate on specific picture aspects during training.

3.1 Introduction

Although previous work (e.g., the original AttnGAN) has set the foundation for this form of synthesis, much more efforts are required to make generating realistic, fine-grained images for practical use. The architecture of the modified AttnGAN is described in this chapter. At the heart of our improvement are three main contributions: multi-scale disc attention is established, a progressive training schema is employed for the generator, and a multi-critic architecture is embedded[10]. At the same time, the progressive training mode makes the generator learn the high-quality features of the low-level feature already at a low resolution high in the early stage, and step-by-step trains the image in a higher resolution stage, which improves both the quality of the convergence and the stability of the training. At the end, the multi-critic model adds different discriminators to each evaluate different characteristic of this generation image, giving the generator more specialized feedback, which enhanced the image fidelity.

3.1.1 Overview of the Proposed Model Architecture

At the heart of a high-quality T2I is the need for a good model architecture. In this chapter, we present the design process of enhancing the AttnGAN model. Leveraging a variety of state-of-the-art techniques, the overall objective is to improve the model's capability of generating realistic and detailed visuals based on natural-language descriptions. These approaches are based on a multi-critic model. In other words, these advances are supposed to address the limitations of the original AttnGAN model, i.e., the requirement for an equilibrium between the fine-grained details of the picture and the general structural coherence. [7] The proposed architecture focuses on enhancing the generator's power to ensure correct representation of both low-level features (such as the textures and fine details) and high-level features (e.g., object layout and scene structure. (generator-for-image) Generative model still lies at center architecture that generate images, and discriminative identify their realness. The primary changes are how we organize the generator and discriminators to provide space in which these advancements are implemented efficiently.

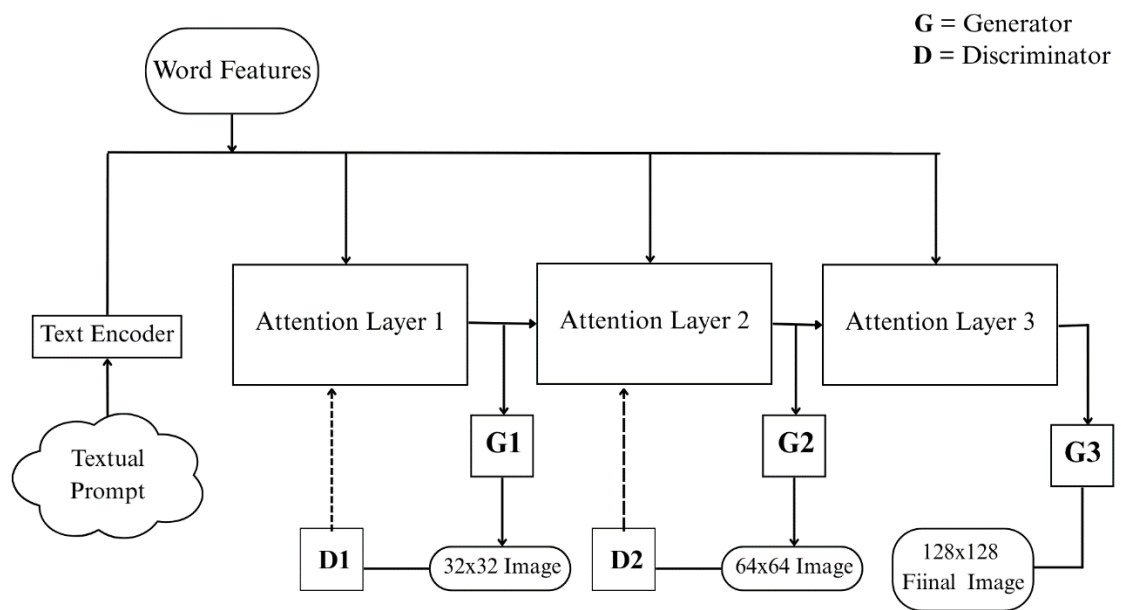


Fig 3.1: Proposed Architecture of our GAN model

Using a progressive training approach, the model learns to generate low-resolution images first and increases the resolution gradually with the training. This staged approach ensures that the generator initially focuses on learning global information and follows by refining the details, and helps the model to make the training process stable [32]. The

multi-critic model provides the generator with a more detailed input for training purposes by utilizing multiple discriminators, each meant to evaluate particular parts of the generated picture (for example texture, object structure or visual quality).

Also, the architecture uses a perceptual loss function instead of pixel-wise loss terms. [7] Through comparing feature maps from the pre-trained networks (e.g., VGG19), our perceptual loss can help the model concentrate on high-level perceptual features, which improves the realism of generated images and makes them closer to the human perception.

The inference of such methods is precisely the objective of the proposed model's architecture, which unifies all of these progress crucial for better text-to-image synthesis. Each part of the proposed architecture will be further described in more detail in the following sections, along with their specific role and the motivation for introducing such model part.

3.1.2 Objectives of Improving the AttnGAN Model

Although AttnGAN made great paces by using attention mechanisms to make text and about aligned to the generated images, there are still a lot of improvements that can be made to generate higher resolution and more detailed images. This part summarizes the major goals of the improved AttnGAN initiatives to remedy its deficiencies and explore the limits of text-to-image synthesis.

1. Enhancing Image Detail and Structure with Multi-Scale Attention:

The first goal is to incorporate a multi-scala attention mechanism into the model. The original AttnGAN was unable to simultaneously model fine-grained details and the global structure of the image, despite attending to the associated text in the generation of the image [31]. The model will have the ability to focus at different spatial resolutions due to the multi-scale attention mechanism, which will improve its ability to generate fine textures as well as the the overall structure of the image. To this end, our contribution is aimed to encourage a more comprehensive and balanced approach to text-to-image synthesis involving rich and contextually plausible visual generation.

2. Stabilizing and Refining Image Generation through Progressive Training:

Progressive training the generator is another major goal. Such a scheme generates

low-quality images and updates them progressively when the model goes through different training stages. The generative process of the progressive training allows the model to learn from global features of the image initially, before focusing on fine-grained details. This staged refinement can contribute to a better training stability, concerned about unstable training and artwork collapse and allowing the model to output higher-resolution images with better perceptual quality. To this end, the objective is proposed to strengthen the capacity of the generator by generating detailed, coherent, and high resolution images from textual descriptions..

3. Providing Granular Feedback with Multi-Critic Models:

Third, we would like to propose a multi-critic model, which uses several discriminators to discriminatively assess characteristics of the generated image. When compared to traditional GAN models which utilized a single discriminator, a multi-critic model can provide detailed feedback on different criterions, like textures, object structures and overall image quality. This goal is to come up with the more diverse and specific feedback to the generator, which enable the generator to refine itself on various aspects, in order to generate more realistic and visually pleasing images. The multi-critic mechanism can effectively overcome the difficulties in generating visually complex and high-quality images according to the ambiguous or unclear input text.

3.2 Model Architecture Design

With the addition of several important components designed for producing high-quality images from text descriptions, the proposed model surpasses the vanilla AttnGAN. The multi-scale attention mechanism, the progressive training strategy and the multi-critic model are the three key improvements in the design. To combat some limitations of AttnGAN, all are carefully selected to achieve the construction of images of higher quality, coherency and resolution in the enriched architecture. The multi-scale attention mechanism is also important to increase the model's ability to focus on the larger structural components of the image as well as finer ones. By incorporating attention at multiple spatial resolutions, the model can focus on finer textures at lower resolutions and larger structures at higher resolutions. A more advanced feature, where the usage of progressive training has been implemented. In this strategy, the generative model first produces low-resolution images and incrementally improves those as the model is trained. This approach helps the generator learn to represent high level features before refining

the details, leading to more stable training as well as higher-quality images as the resolution increases. [6] The training is progressive, that is, the generation model scavenges common issues of mode collapse and progressively generates more coherent and higher-quality images.

In addition to these developments, the architecture consists of a multi-critic model that consists of multiple discriminators to assess different characteristics of the generated image. As opposed to classic frameworks with an unified discriminator, multi-critic designs special discriminators to emphasize different levels of details, such as texture, structure, and general visual fidelity. This finer feedback gives the generator more information about the specific ways it needs to get better, and produces more detailed and realistic samples as a result.

3.2.1 Multi-scale Attention Mechanism

One of the main contributions in the proposed model architecture is the multi-scale attention mechanism, which aims to alleviate one of the common challenges in text-to-image generation: generating precise and coherent images. In classical models, an attention mechanism is commonly used on one resolution level, causing it to generate too detailed or too general structure. The model can attend on different resolutions simultaneously, due to the multi-scale attention mechanism, which overcomes this limitation and enables it to capture both the large-scale object structures and detailed textures in the same image.

Mathematically, the multi-scale attention mechanism works by computing attention weights for each spatial resolution. The attention map[15] at each scale k is computed as follows:

$$A_k = \sigma(\text{Conv}_k(T \odot I_k))$$

Where:

- Conv_k is the convolution operation at scale k , applied to the element-wise product $T \odot I_k$ which represents the interaction between the textual features T and the image features I_k .

- $\sigma(\cdot)$ denotes the sigmoid activation function, ensuring that the attention weights lie between 0 and 1, allowing the model to determine how much attention to give to different regions of the image.
- $\odot$ represents element-wise multiplication, ensuring that the attention mechanism is applied based on both the text and the current image features.

These attention maps are then applied to the image features at each resolution to emphasize the relevant regions for the generator. The final image is generated by aggregating the information from multiple scales. The output of the generator can be expressed as:

$$I_{final} = \sum_k A_k \odot I_k$$

Where I_{final} is the final generated image, which is a combination of features in separate attention maps. The model is therefore not only able to parallel learn the details of the locals along with the general pattern of the image and, as a consequence, provide a more coherent and detailed output.[13]

The primary advantage of the multi-scale attention mechanism can be discussed as its ability to combine the global shape of pictures with their fine-grained details. Since written descriptions may cover both fine detail (such as color and texture) and high-level context (such as relations between items and the layout of space), this is particularly important in text-to-image synthesis. The multi-scale attention approach ensures that the resultant images are both contextually very accurate as well as visually believable by allowing model to focus both aspects very closely in line with the textual description offered.[14]

Nevertheless, multi-scale attention mechanism has its own barriers too. One of the main issues can be seen in the computational elaborateness of computing multiple attention layers at diverse resolutions.[13]The rise in computing requirements of the model with the number of scales increase can cause longer training times and memory usage. Also, when the attention mechanism is not properly regularized, it is possible to suffer overfitting that can lead to the negative generalizability on an unseen data set, although, the multi-scale approach enhances the ability of the model to target different aspects of a picture..

3.2.2 Progressive Training Approach

The second major enhancement to the proposed model architecture is progressive training method. This is intended to improve the overall quality of pictures generated and the consistency of training even during the creation of images of high resolution. Because of issues such as mode collapse, difficulty in training stability, and ability to sample fine-grained details, training on high-resolution images can be a problem to traditional GANs. The progressive training technique, which gradually increases the resolution of the pictures created during the training process, will solve these challenges.

Under this model, the generator is first made to generate only low-resolution images in an effort to formulate general structure of the image and its global contents. After the generator has been trained at a level of this lower resolution, the model is progressively exposed to increased resolution of images in later steps of training.[12]Working with low-resolution images at the beginning helps the generator learn the rough layout of the scene, i.e. big objects, placements and design of the entire scene. After learning such global characteristics, the model continues refining the image by adding more detailed characteristics at higher resolutions.

Mathematically, the sequential learning procedure may be considered as the training to generate an image I_r at resolution r of the generator G . The learning starts off at the lowest resolution r_{low} and progressively through stages towards the desired high resolution. The loss function at each stage r is varies to learn the features that are relevant at that resolution beginning with global features of layout and next learning more localized features.

This enables the generator to learn, initially, to generate crude representations of the image at lower resolutions and slowly enhance that output with learning of detail textures and small objects details, as more detail is added to the image. Each stage loss is well weighted and the initial training of the network focuses on the big structure, whereas the later stages are concerned with optimizing minutiae.

A common issue with GAN training, mode collapse, is also reduced in this progressive training. Mode collapse occurs when the generator produces a single few pictures and often the consequent output is not very diverse. Partly, the training procedure of gradually

training the model also enhances the variety and quality of generated images, as it motivates the generator to learn the entire range of image characteristics, both general structure and complex textures. Moreover, such training procedure stabilizes the learning process. Training The usual training procedure in traditional GANs is training both generator and discriminator together, which can produce instability, particularly with high-resolution images[8]. The progressive training scheme has the advantage that earlier phases use lower-resolution images, allowing the model opportunity to become familiar with the large-scale structure of the image first, and forcing the model to master the high-level structure of the image before learning the richness of the fine details. This progressive learning allows the generator to converge much easier and it is less likely to oscillate and fail to converge in a training phase.

But, along with the positive aspects of the progressive training approach, there are also difficulties with this approach. The main problem is in deciding the best time to amplify the resolution in the training. When increasing the resolution in too rapid a manner, the generator might find it hard to integrate the requisite information which results in poorer resolution of the images.

3.2.3 Multi-critic Model Integration

Introduction of a multi-critic model is actually an impactful improvement of the typical GAN model. Although this has been effective, the use of the single discriminator is a limitation to a deep feedback that the generator could have gotten during generation of complex and high fidelity images [37]. Multi-critic model circumvents this shortcoming by putting in place various discriminators having individualized expertise at assessing specific analysts of the images.

In multi-critic configuration, discriminators are intended to score distinct characteristics of the image produced. One of the discriminators could be concerned with the texture of the image so that the fine details such as the surface patterns and the material properties could be represented fairly. The other discriminator would perhaps consider the structure of the picture, whether the objects have been arranged at the right position to each other as well as whether the entire scene arrangement makes sense. The overall visual quality, which can be based on such areas as lighting, contrast, and other aspects of the picture with regards to aesthetics appeal, could be another discriminator. Breaking the evaluation

task into specific tasks, individual discriminators can offer more specific feedback in order to make the generator improve particular parts of the image.[11]

Mathematically, the multi-critic model can be represented as a series of discriminators $D_1, D_2, \dots, D_n$ where each discriminator D_i focuses on a particular feature of the image. The total discriminator loss is calculated as:

$$L_{Total} = \sum_{\{i=1\}_{i\{n\}}\lambda} L_{\{D_i\}}$$

Where:

- L_{D_i} represents the loss function for discriminator D_i , which evaluates a specific aspect of the generated image.
- λ_i is a weighting factor that controls the importance of each discriminator's feedback in the overall loss function.

The generator is updated based on the total loss L_{total} , ensuring that the feedback from all discriminators is considered.

Separating the evaluation of the various features of the image, each discriminator may be able to give more precise information, and this information helps the generator to concentrate on the optimization of these specific features. This makes the images more coherent and good looking. As an example, whereas one discriminator can say that the image has realistic textures, another can take care of the spatial relationships between objects and in such a way produce more realistic images in general.

Nevertheless, the multi-critic model also has the drawbacks. Among the key difficulties is the need to understand how to equalize contributions of every discriminator. When the feedback of one discriminator is weighted too highly, this risk is run of dominating the training process, leading to the generator not being able to optimize other parts of the image. But on the other hand, an over-balanced distribution of the feedback might hinder the model to make one feature noticeably good enough. The weighting factors Over preparations are necessary to make sure that the weights applied by each discriminator are well adjusted so that none dominates over the other.[16]

As well, a model that uses many discriminators makes the model computationally more complex. The number of parameters required by each of the discriminators determines the size of the model and decreases the time of training.

3.3 Data Preparation and Training

The proposed model requires a well prepared set of data and a well laid training procedure in order to be effectively trained. It is described how data have been collected, preprocessed, and what steps were carried out to train the model and provide an adequate level of convergence and a quality of generated images.

Data preprocessing is the initial important step in training the model as it was deemed crucial to collect and prepare the data. In the case of the text-to-image synthesis, we need to have a dataset of paired textual descriptions and the appropriate images. This study makes use of such common benchmark datasets as the CUB 200 2011, which consisted of a rich diversity of images with detailed textual descriptions [17].

Upon gathering the data, the process of data preprocessing comes in gear, to ensure that images and textual description is in appropriate training context. Preprocessing entails a number of processes such as normalisation of images to make the pixel value and dimensions consistent, and tokenization of the text descriptions to put them into a numerical form to enable the model to process it. Textual information is commonly represented using word embeddings which for word2vec represent each word in the description as a dense vector in a continuum space.

As the data is ready, it is now time to do the training. The training designation is aimed to optimum the model parameters in a way that the generator creates quality of images that parallels the supplied textual representations whereas the discriminator performs well to discriminate between a real and a fabrication image[18]. The training is adversarial, and the generator, discriminator is trained in a minimax game. The generator will tend to minimize the loss function and the discriminator tries to maximize the capacity of being able to tell the difference between the actual and the generated images.

To reliably renew the weight of the model, it is necessary to use the optimizers during the training process. The Adam optimization algorithm is the common optimization technique with known flexible learning rates being used in the study. Since it is a combination of Adagrad and RMSprop, Adam is perfectly applicable to train models with

sparse gradients and large datasets. The high learning rate is optimized the fine tuning model for high quality picture generation by fine tuning the learning rate to render a smooth convergence, as well as fine tuning the learning rate decay during training. The main loss functions include:

1. **Adversarial Loss:** The discriminator tries to discern between actual and false pictures, and the generator tries to trick the discriminator[19]. This is the fundamental loss function of the GAN architecture.

The adversarial loss for the generator GGG and discriminator DDD can be formulated as:

$$\begin{aligned} L_{adv} &= E[\log D(x)] + E[\log(1 - D(G(z)))] L_{adv} \\ &= E[\log D(x)] + E[\log(1 - D(G(z)))] L_{adv} \end{aligned}$$

where x is a real image, z is an input text or random noise vector, and $D(G(z))$ is the discriminator's output for a generated image.

2. **Perceptual Loss:** The absence of realistic images is often seen when traditional pixel-wise losses are used since they do not measure the perceptual quality of the image[20]. This is solved by incorporating perceptual loss through which the difference in the high-level characteristics between generated and real pictures is quantified. The feature maps are compared through passing through a pre-trained network to compute the perceptual loss:

$$\begin{aligned} L_{perceptual} &= \sum_l \|\phi_l(I_{real}) - \phi_l(G(z))\|_2^2 L_{\{perceptual\}} \\ &= \sum_{\{l\}} \|\phi_l(I_{real}) - \phi_l(G(z))\|_2^{2L_{perceptual}} \end{aligned}$$

where ϕ represents the feature maps at layer l of the VGG network, and I_{real} and $G(z)$ are the real and generated images, respectively. This loss helps the generator produce images that are perceptually more realistic and aligned with human visual perception.

3. **Reconstruction Loss:** In some implementations reconstruction loss is used to ensure that the generator produces not just lifelike images but also that the image produced should be very similar to the textual description being given. Such loss can be

computed by comparing the feature embeddings between the created image with textual embeddings of the description.[22]

3.4 Evaluation of Model Performance

The performance of the proposed model is measured both by quantitative measures of assessment and by qualitative methods of evaluation. These evaluation techniques give a complete account of the ability of the model to generate realistic and quality images using written descriptions. By merging numerical measurements with human estimates, the output of the model will always be perceived accurately and consistent with the provided text. Inception v3 network is a pre-trained image categorization model [21] which is the base of the IS. It tests two major aspects of the quality and diversity of the images produced. The IS is computed by using the Kullback-Leibler divergence between the label distribution of marginal labels of the predictions made by the Inception network and the conditional labels of the generated pictures.

The other criterion that can be used to measure the quality of the generated pictures is FID which compares the distribution of feature extracted in actual images and features extracted based on the Inception network. The FID estimates the divergence between distribution of feature vectors generated and real images by focusing on mean and covariance of feature vectors [23]. The lower FID score further implies more quality and the sense of reality since it indicates that made images are more representative of the original ones.

Both the IS and the FID provide essential details with regard to the performance of the vehicle. The FID compares the distribution of generated and actual pictures in its entirety and gives more detailed conclusion on image quality compared to the IS that only considers variety and quality of individual generated pictures [24].

Although quantitative measurements have their importance (objective measurements), qualitative methods of evaluating images are also beneficial and must not be overlooked. The variants are based on human judgment in case of the visual coherence, realism, and correspondence of the generated images with the provided textual descriptions.[50]

The qualitative evaluation can be accomplished via user study where people can be shown pairs of generated images along with text descriptions accompanying them. Thereafter, the evaluators are requested to rate the images according to various elements, including:

- **Relevance:** How closely the description and the image match.
- **Realism:** How realistic and visually appealing the image looks.
- **Coherence:** How well the objects and elements in the image are organized and structured.
- **Detail:** The level of detail present in the image, especially for textures, lighting, and other subtle elements.

Another criterion by which the image-text match can be assessed is the correlations between some fine specific details in the picture and the details referring to the text. As an example, when the text says: a red apple with a shiny surface, the judges would decide whether the apple in the produced scene is really red, shiny and proportionally correct.[23]

In other instances, human evaluators may also be asked to compare the images generated in the experiment with other actual images with descriptions on the same to determine whether there is possible difference between the two. There is a reciprocity in the quantitative and qualitative assessments. The quantitative measures give objective numerical data of the output of the model in terms of success, the qualitative measures however ensure that the produced images are similar to what a human would do in terms of visual and textual congruency..

3.5 Summary

Through this chapter, the proposed model architecture was described in terms of design and components that would better synthesize. The main innovations implemented are the multi-scale attention mechanism, gradual method of training and combining multiple critic models. All of these elements were selected with the goal of overcoming certain weaknesses of the original AttnGAN model and taking into account the fact that the new architecture should be able to take highly realistic images based on the given textual descriptions. This allows the model to produce high-textured and structurally consistent images, raising the level of realism as well as the precision of the created content. Progressive training strategy stabilizes training by slowly ramping up image resolution as part of the training process, so that instead of immediately attempting to learn about

the fine details, the generator is initially concerned with the global features. Multi-critic model proposes multiple discriminators, each of which looks at a particular part of the generated image, e.g. texture, structure, general overall image quality, etc. thereby giving more discretely detailed feedback to the generator, and allowing the generator to refine multiple aspects of the image at the same time..

Chapter 4

Socio-Economic Impact and Sustainability

The future of enhancing sophisticated AI, including a more effective AttnGAN, to assist in developing text-to-image synthesis techniques, has the immense implication not only in relation to the technological advancement but also in regards to socio-economic and sustainability. Although the main objective of enhancing the process of T2I is to generate more realistic and realistic imaging systems, the wider reflection of such developments should also be addressed. The purpose of the current chapter is to investigate socio-economic opportunities and disadvantages of the proposed model.

There would be major economic changes due to the interaction of AI and machine learning in innovative areas like art, design, marketing, and entertainment. It could help to speed up creative processes, lower the expenses, and simplify the process of prototyping and creation of content. Nevertheless, such developments are followed by challenges that have to be settled, namely, by the replacement of modern jobs and the ethical importance of AI-created material. The effects of AI on employment could occur in industries that require the services of human designers and artists as AI becomes the force behind automation in the creative process. These challenges will be addressed with increased importance on policy-making and retraining and adaptation of the workforce.

4.1 Introduction

The developments in the field of text-to-image synthesis, are the step towards the different things, such as entertainment and advertising, e-commerce, education, etc. The technologies also give new frontiers of innovation and creativity, but also pose concerns about their overall effects on the society and economy. Socio-economic impacts of AI are vivid and include both favorable and adverse outcomes. On the one hand, AI-based tools have a potential to increase productivity, reduce expenses, and make high-quality design and content creation democratic. Conversely, with the emergence of automation in the art and design sectors, conventional employment sectors could become one of the victims of this revolution, in particular those in which creative humans have been and continue to be applied to design, art, and content production. Also, in case of the development of AI technologies, such as text-to-image models, it is a growing factor to consider from the

point of view of sustainability. It is therefore of crucial importance to discuss the sustainability of these technologies through the prism of long-term effects on the environment with a narrower spectrum of minimizing its carbon footprint and fostering sustainable AI practices.

4.2 Impact of the Thesis on Societal, Health, Safety, Legal, and Cultural Issues

On the one hand, these developments democratize expression of creativity as they allow individuals and organizations to create high-quality images using limited resources. It can improve accessibility, especially to persons with low access to traditional tools or design skills. Nevertheless, there are ethical and legal issues concerning these technologies, too. With the increase in the number of AI-generated content, concerns about the copyright of ideas, intellectual property, and authorship become relevant.* The issue of ownership of the works created using AI and the creation of proper legal grounds to ensure the protection of the creators is an important topic to maintain a fair distribution and use of the AI produced materials and to stop unfair, inappropriate use of such materials. Health and safetywise, the capacity of AI to produce hyper-realistic pictures might lead to misguiding or even detrimental material, e.g., deepfakes or fake medical pictures, which can have severe outcomes in any field, such as healthcare or even the wellbeing of the community. Its social effect is also significant, but it can result in the loss of jobs in such industries as design, advertising, and media creation that require the involvement of human creativity. This will need proper planning, on how to retrain the workforce and flexible policies to reduce the adverse effect. As far as the cultural aspect is concerned, even though AI presents new opportunities of creative expression, it expands the limits of art and creativity.

Chapter 5

Addressing Complex Engineering Problems and Activities

There is a set of multi-faceted engineering issues that should be solved to implement advanced AI models. These issues vary between the scalability, efficiency of the models to the optimization of the models in terms of its computational capability and its impact on the environment. Also, to ensure that these models make sense when applied to real life applications, either in entertainment, in the design or medical fields, their requirements need to be critically looked into as far as their operations are concerned in relation to: system integration, data management and the relationship they have with their users.

5.1 Introduction

These challenges are twofold: while the first ones are related to the technical aspects of designing and training a model, the second ones concern the implementations and scaling issues to consider when applying such models in production. To tackle these challenges, engineers have to make the model optimal in terms of both performance and efficiency, while making sure the model can be benefited by real-world use-cases. The complexities of actually deploying these types of models into the wild also need to be considered. This not only requires the models to be of sufficiently high quality, but also the model's behaviour to fit into the system architecture that allows for scalable deployment. Solving all of these engineering difficulties is required to make state-of-the-art models such as AttnGAN not just possible in a research scope, but also useful in commercial applications. [24]

This chapter also investigates engineering challenges involved in applications of AI models such as AttnGAN in various sectors including, scalability, computational efficiency. Large scale data processing and fast and accurate inference are the key rules of the game, and justify more and more advanced optimization strategies. Additionally,

it is very important that such models can be applied in real-time scenarios such as existing production lines with limited computation and without much degradation of the output quality.

5.2 Addressing Complex Engineering Problems

One of the major challenges is how to improve the computational efficiency of the model. Text-to-image models are computationally expensive, notably during the training phase. Highly performance properties should be preserved, while the resources are efficiently managed. This frequently means taking advantage of parallel computing and distributed processing to accelerate training and inference. Another critical problem is the scalability of the model. When the model becomes more complex and the data more extensive, this scaling ability is essential. Methods like model pruning, quantization and hardware acceleration mitigate these issues by shrinking the model size and computational overhead while maintaining performance.

And they have to be robust, too. The model need be robust to various types of input data such as noisy, partially-existent or ambiguous natural language descriptions, while remaining capable of producing high quality images. This issue is mitigated by using advanced data augmentation methods and regular updates of the model architecture.

Table 5.1: Complex Engineering Problem and their Corresponding Solutions

Engineering Problem	Description	Solution
Computational Efficiency	Optimizing the model to manage computational resources effectively, especially during training and inference.	Using parallel computing, distributed processing, model pruning, and hardware acceleration like GPUs or TPUs.[27]
Scalability	Ensuring that the model can scale to handle large datasets and more complex tasks while maintaining performance.	Implementing model pruning, quantization, and optimizing hardware usage to ensure scalability.
Robustness	Enhancing the model's capacity to produce high-quality pictures even when dealing with varied, noisy, or missing input.	The model architecture and applying data augmentation strategies to increase generalization and robustness.
Integration with Existing Systems	Ensuring compatibility with existing software platforms, handling data management, and	Integrating the model with appropriate APIs and ensuring it is capable of

	maintaining real-time processing capabilities.	managing large-scale data efficiently for seamless integration.
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5.3 Address Complex Engineering Activities

It deals not only with the position of engineering problems but also with the solving of complex engineering tasks that are indispensable in order to adopt used by advanced models in practical applications. Such tasks involve streamlining the the training pipeline, enhancing support for large datasets, and deploying the model in production. The training pipeline needs to be high-performance and high-quality, allowing it to process large-scale data effectively and efficiently without the sacrifice of the model's effectiveness. It means we use distributed training, gradient accumulation, and even hardware acceleration (like GPUs and TPUs) to speed up the process. Efficient data storage, retrieval and preprocessing would be needed to manage the volume and variety of the data without becoming bottlenecks during training.

Finally, the modeling must be integrated into production systems which includes system architecture design and integration with existing platform. This involves working with software developers to create scalable and robust APIs, implement real-time data input, and deploy the model in a resource efficient way.

Table 5.2: Complex Engineering Activities and Approaches

Engineering Activity	Description	Approach	Tools Used	Challenges
Optimizing Training Pipeline	Speed and efficiency optimization	Distributed training, parallel computing	GPUs, TPUs, Horovod	Balancing efficiency with quality
Managing Large-Scale Datasets	Handling large data volumes	Data augmentation, parallel loading	Apache Spark, TensorFlow	Data storage and retrieval
Integration into Production	Seamless system integration	APIs, cloud-based solutions	Docker, AWS/GCP	System compatibility
Model Optimization	Fine-tuning model parameters	Grid search, transfer learning	TensorFlow, PyTorch	Ensuring generalization
Hyperparameter Tuning	Optimizing hyperparameters	Cross-validation, grid search	Scikit-learn, Optuna	Managing large search spaces

Performance Monitoring	Monitoring and debugging	Log monitoring, profiling	TensorBoard , PyCharm	Identifying bottlenecks
Scalability Testing	Stress testing for scalability	Cloud infrastructure, load testing	AWS, GCP, Docker	Resource management

5.4 Summary

This chapter presented the intricate engineering challenges and the tasks for deploying and tuning the model for real-world applications. It brought up several important issues, including making the training pipeline more efficient, dealing with large-scale datasets, and seamlessly moving models to a production environment. Several engineering activities, such as modeling refinement, hyperparameter tuning, real-time performance tracking, and frameworks' scaling, were described along with the methods, tools and issues in their implementation. Overcoming these issues guarantees a successful implementation of the model and improves its performance and applicability in many scenarios.

Chapter 6

Thesis Management and Finance

6.1 Introduction

This chapter describes the general outline of the thesis based on the management plans, timeline, as well as the financial budget. Improving generation quality The work will require improvements upon the current AttnGAN model, in order to generate more realistic images from descriptions. To achieve successful implementation of this investigation, meticulous planning, scheduling and budgeting are mandatory. Proper management of technical and financial aspects for the project will help it be completed within the project scope and budget.

The thesis will involve the combination of a number of engineering and computational tasks: model design, data refinement, training and performance assessment. A comprehensive financial plan is required to properly budget resources, including computational costs and data collection costs. This chapter describes the overall management organization, the timetable and the financial planning of the project that form the baseline for the implementation of the research within time and cost schedule.

6.2 Management of the Thesis

The research will have different sub-studiespace, each focusing on separate dimensions of the model creation. It consists of these stages which are:thesis writing service Get the Best Thesis Writing Service Online literature review, data collection, and presentation and documentation. Progress checks and milestones will be established throughout the project to monitor progress, revise workplans, and to ensure the project is kept on track.

It offers proper supervision and professional assistance while working on the project. Contact with the project supervisor and faculty members must be maintained, so that the student remains on course, problems are solved and the thesis at the time of submission meets what is generally accepted as good quality work at the School.

6.2.1 Time Frame

A comprehensive schedule of work has been established for the thesis to ensure the tasks be done in the specified time. The project will cover the two academic semesters and have well-defined milestones and deliverables in each phase. The work plan is structured to be consistent with the pace of the institutional work, so that both theoretical studies and model development is achieved in an optimal manner. For each stage, there are key milestone and review points to allow iterative development of the model, and regular check-ins to evaluate progress and make course corrections as necessary.

Table 6.1: Time Frame of Our Thesis

Phase	Duration (Semester)
Literature Review, Problem Identification, Data Collection	4th Year Odd Semester
Model Development (GAN Enhancements, Training, Data Preprocessing)	4th Year Even Semester
Evaluation (Quantitative Metrics, Qualitative Evaluation, Final Report)	End of Even Semester

6.3 Overall Budget

This thesis is materialise, credit-financial wise. The budget is allocated across categories to make sure that whatever is needed is there, within the budget. The main expenses are computational costs (e.g., cloud services, GPU renting), software licenses, data purchase and some other expenses in materials for research and documents.

Table 6.2: Overall Budget

Sl.	Item	Specification	Justification	Quantity	Unit Price (BDT)	Total Price (BDT)
1.	Processing Memory	RAM: 64 GB	Needed for training large GAN models	1	32,000	32,000
2.	GPU	NVIDIA RTX 4090	Best-in-class performance for GAN	1	220,000	220,000

			training			
3.	VRAM	24 GB (Part of RTX 4090 GPU)	Essential for high-resolution image synthesis	1	9,000	9,000
4.	Storage	NVMe SSD (1TB)	Fast access to large datasets	1	-	-
5.	CPU	AMD Ryzen 9 7950X	Required for efficient pre-processing tasks	1	60,000	60,000
6.	Cloud Computing Resource (If required)	Google Cloud TPU v4	High-speed tensor processing for GAN training	36	300/hour	10,000
	Total (BDT)					331,000

In words: Three lakh thirty-one thousand Bangladeshi Taka.

6.4 Software Budget

Besides the hardware requirements, the following software packages and components are indispensable to realize and run the improved AttnGAN model. The software budget includes licensing costs for deep learning frameworks, cloud services, and all other required tools to support data pre-processing, model generation, and assessment. Main deep learning platforms used: TensorFlow and PyTorch Two popular open-source deep learning packages widely known in the research community for building and training deep neural networks.

During the testing, trained models can be tested using CPU, while CUDA Toolkit can be used for GPU acceleration on the intensive training. In addition to project services being delivered on Google Cloud's TPU v4 and cloud storage solutions, large model training will be performed with high speed processing power needed to manage the massive datasets and computational needs of the research.

Other auxiliary software tools like Jupyter Notebooks, Scikit-learn, and OpenCV will be utilized throughout the process for data preprocessing, visualization and benchmarking. These tools aid the development process, helping with data processing, model evaluation, results export for further analysis etc..

Table 6.3: Software Budget and Specification

Sl.	Item	Specification	Justification	Quantity	Unit Price (BDT)	Total Price
1	Deep Learning Framework (TensorFlow)	Open-source deep learning library for model training	Building and training deep neural networks	1	-	Free
2	PyTorch	Open-source machine learning library	Implementing the GAN model and model evaluation	1	-	Free
3	CUDA	CUDA Toolkit for GPU Acceleration	GPU-based computation during model training	1	-	Free
4	Google Cloud Services	Google Cloud TPU v4 and storage	Cloud-based computing resource for large-scale model training	1	-	Free
5	Miscellaneous Software Tools	Jupyter Notebooks, Scikit-learn, OpenCV, etc.	For data preprocessing, visualization, and evaluation	1	-	Free
	Total (BDT)					0

By relying on open-source software, the project minimizes costs while still utilizing state-of-the-art tools and services necessary for advancing the AttnGAN model.

6.5 Summary

This chapter covered the key management, time and money planning needed to ensure the successful completion of the thesis work, which is dedicated to extending the AttnGAN model for high-resolution text-to-image synthesis. The proposed approach is organized as a multistage research project, with specific tasks and deliverables in each stage, which will considerably facilitate systematic and well-structured model-theoretical framework development, as well as data pre-processing, training and validation. A clear timetable has been set for two academic semesters and project checkpoints to track and

enhance. Frequent contact with the thesis advisor and professors keep the project on the cutting edge of academia, as well as address any spur-of-the-moment problems.

The total cost has been well-structured with respect to computational resources, software tools and other essential overheads. Details are given on hardware requirements such as high-performance GPUs and cloud-computing resources, which facilitate the required infrastructure for iterative efficient modelling. The software budget (including scheduling algorithms, deep learning frameworks, and cloud services) is designed to rely on open source tools with minimal fees to enable minimum cost and maximum utility.

Chapter 7

Design Methodology and Implementation

7.1 Introduction

This is a significant step where we move from model design and theoretical examination to implementation, that is, training mechanism, of our model of the proposed system for improved AttnGAN model for high-resolution T2I generation. The CUB_200_2011 bird dataset containing images of 200 different bird species, is also extensively employed in this work. Several textual captions are provided for each bird image, describing color, shape, and posture. These descriptions are used as input to the AttnGAN model that attempts to create images that capture the essence of the textual descriptions. [26]

The training of the model was performed in Google Colab on T4 GPUs for 600 epochs due to the model's complexity and the dimension of the dataset, with resumable training strategy for interrupted training. The next parts of this chapter will explain our design methodology and components, optimization mechanisms and experimental arrangements, with a greater emphasis on the data-handling pipeline and the machine-learning framework settings. The goal is to introduce a straightforward and effective pipeline where model training, evaluation, and testing are unified to obtain the desired enhancement in the quality of synthesized images.

7.2 Detailed of Design Methodology

This section presents the design and methodology of the AttnGAN model and the detailed procedure to generate and improve it. The method is data-oriented and iterative, concentrating on the model performance enhancement with the proven method to be able to improve the performance, i.e., structured data preprocessing, model building, and performance assessment that are semi regulated. The purpose of this method is to synthesize realistic images from text descriptions, exploiting advanced techniques including multi-scale attention, progressive training, and multi-critic fusion.

7.2.1 Data Collection and Preprocessing

The model for generating available bird images starts with the CUB_200_2011 bird dataset, which provides for each of its 200 bird species 10 textual captions. The diversity and rich descriptions of the dataset make it a good candidate for the high-fidelity text-to-image synthesis. The data pre-processing pipeline take in the following steps.

- **Caption Embedding:** Ten captions corresponding to an image were tokenized and pre-trained word level embedding (Word2Vec) was used to encode vector representations for each word. This makes the relationships between the words and assist in the model generating meaning from the text. [27]
- **Creation of Pickle Files:** After the creation of captions, they were stored in pickle format for ease of storage and access. This approach enables the embeddings to be efficiently loaded at run time (e.g., during model training and testing), minimizes memory consumption, and speeds up both learning and inference process. [41]
- **Preprocessing images:** Images were resized to a fixed resolution in order to keep consistency across the set. We employed normalization, i.e., the scaling of pixel values to the interval [0, 1], to improve model convergence. [30]
- **Data Augmentation:** To mitigate overfitting and increase the variety of the dataset we adopt data augmentation techniques such as random flipping, rotation, and cropping, especially for those rare bird species.
- **Data Split:** a train, validation and test sets of dataset were split. The training set consisted of 80% of the photos and the remaining 20% of them were divided in testing (10%) and validation (10%). This paragraph ensured that the model was tested on unknown data points to test its generalization capabilities as well as a good cross section of examples that it had been trained under.

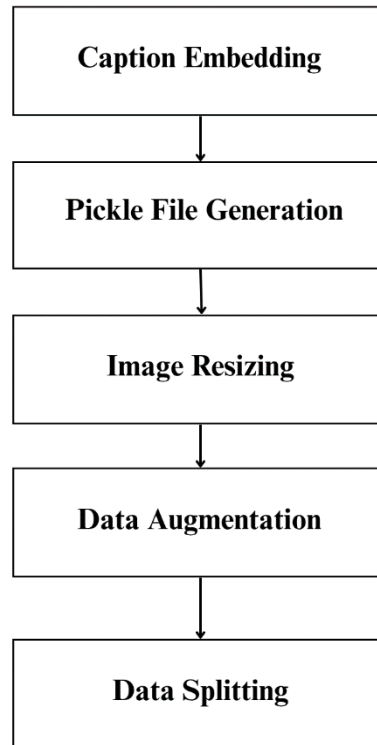


Fig 7.1: Data Preprocessing Steps

7.2.2 Model Architecture Design

The AttnGAN model was improved to improve the realism and quality of the generated images. The architectural changes aim at enhancing the attention mechanism of the model on various regions of the image and at better conditioning of the generated images over time.

- **Multi-scale Attention Mechanism:** The multi-scale attention mechanism was used to enable the model to attend to different spatial regions with various scales. This allows training of the model, including the virtual camera or the visual cues to the support of such as texture and object shapes, being important for high quality image generation. [34]
- **Progressive Training Strategy:** Type II.CMP's model was trained progressively (from low to high resolution), and the process of model update was divided into stages to learn to generate photos of graduated resolutions. This allows the model to learn coarse features first, where they can be refined subsequently in high resolution stage, better preserving details with sharper images. [25]

- **Multi-critic Model Integration:** We used a multi-critic approach, such that multiple discriminators (critics) were used to evaluate a different aspect of the generated image. Each critic concentrated on assessing a different property of the image, such as a graphical realism, object consistency, and background detail. Multiple critic feedback increases the general quality of produced pictures, because a more complete evaluation is carried out during training..

7.2.3 Training and Experiment Setup

Training was carried out in a cloud-based environment using Google Colab with T4 GPUs, offering the computational resources needed for large-scale model training. The following parameters were used for training:

- **Epochs:** 600
- **Batch Size:** 64
- **Image Size:** 128x128
- **Learning Rate:** 0.0002
- **Optimizer:** Adam Optimizer
- **Resumable Training:** To handle potential interruptions in training, resumable training was implemented. This allowed the training to resume from the last saved checkpoint without restarting the entire process. This feature was particularly useful when working with extended training sessions that span several days.

7.2.4 Evaluation

The model's performance was evaluated using both quantitative and qualitative methods:

Measurable Evaluation Metrics:

- **Inception Score (IS):** Measures the diversity and quality of generated photos.
- **Frechet Inception Distance (FID):** Measures similarity via the comparison of the distributions of generated and real images.
- **Perceptual Loss:** A loss function that encourages the generated image to capture the perceptual similarity of the real image by exploiting the knowledge of human perception.

Qualitative Evaluation:

- **Visual Inspection:** The generated images were visually inspected manually to assess the quality, consistency with the text descriptions, and fidelity to the input text descriptions.
- **Diversity:** The diversity of generated images from a same caption was checked to see the ability of model to produce more than a single unique image for a single input sentence.

7.2.5 Hyperparameter Tuning and Optimization

To achieve optimal performance, several hyperparameters were fine-tuned:

- **Learning Rate:** Experimented with different learning rates of 0.0001, 0.0002, and 0.0005 to find the optimal convergence rate.
- **Batch Size:** The batch size was tested at different values (32, 64, 128) to see the effect on model training speed and stability.
- **Attention Layers:** The total number of attention layers in the model were varied to find the balance between model complexity and strongest performance. The number of layers was also adjusted, from 2 to 4, to observe their effects on image generation quality.
- **Number of Critic Networks:** We estimated the best number of critic networks for evaluating the quality of images during learning. After some experimentation, it was found that the best performance was obtained using 3 critics.

7.2.6 Model Testing and Validation

The model was experimented under several settings to demonstrate its capacity to generate the realistic and versatile images among bird categories. The experimental conditions were as follows:

- **Normal Operation:** In this case, we tested model on full CUB_200_2011 dataset and model has generated images for all 200 bird categories.
- **Low Resolution Inputs:** An evaluation of the model's performance in producing high-quality images from low-resolution inputs was performed to mimic real-world scenarios where fine-grain details are scarce.

- **Semantic Consistency:** We evaluated the ability of the model to generate images that accurately reflect the content of the text descriptions using the semantic consistency metric.

7.3 Selection of Parameters

Note that the effectiveness of the augmented AttnGAN in learning to generate high-quality photo-realistic images conditioned on text descriptions is sensitive to the choices of multiple parameters and modules in both design and deployment stages. It is these parameters which affect model performance, training speed and image quality.

The first key parameter was the size and the composition of the dataset used for training of the model. We selected the CUB_200_2011 dataset as it is rich in details with high diversity and provides 200 bird species with 10 captions per image. The textual descriptions on the provided captions were detailed and diverse, which is important for training a model for text-to-image generation. The dataset was divided into training (80%), validation (10%), and testing (10%) sets, thus ensuring that the model could be evaluated on unseen data to avoid over-fitting and generalize.

For data pre-processing, various preprocessing steps were employed to preprocess images and captions to make both them suitable for the model. Captions were then enriched with several pre-trained embedding models, highly used to learn semantic relationships between the words. These embeddings helped the model build a better understanding of the meanings and relations of words present in the image captions, thereby generating more accurate and believable images. Moreover, pixel normalization was performed to normalize pixel values to $[0, 1]$, which benefits to the effective learning of the model, and the risk of overfitting was minimized. As for model architecture, there are some key design choices for improving the image generation process. The attention mechanism in multiple scales was an essential extension to the source AttnGAN. It is a crucial feature for accurate images since it enables the model to attend the important visual features instead of paying attention to the distractors. Another notable design choice was the progressive training strategy, in which the model first generates low-

resolution image and gradually up-scales until the end of training. The model construction of the multi-critic model was another important aspect. Differing from the original AttnGAN that employed only one discriminator to evaluate the generated images, our enhanced AttnGAN utilized various discriminators to evaluate different parts of the generated image. For instance, a critic may look at the global photo-realism of the picture, another one may evaluate the robustness of the objects or of the background details. Applying multiple critics gives more elaborated feedback and leads to better images and better fit during training.

We selected few parameters for optimization of the learning process in the training configuration. We used a batch size of 64 which provides a tradeoff between memory and computational capacity. Too large a batch size could overflow the memory, while excessively small would assign lower training acceleration [28]. After testing with different learning rates, an initial learning rate = 0.0002 was found to be effective in terms of significantly fast model convergence yet avoiding overshooting. This optimizer is called Adam and it is considered to be very well suited for training and has a built-in mechanism to adjust the global learning rate, which usually leads to a faster convergence and better performance, especially in adversary models as GANs. They also make the training process resumable, where even if the training was interrupted, the model can be reloaded from the last checkpoint, which is helpful in long training sessions to avoid loss of data.

The Hyperparameter tuning also played a role in enhancing the performance of the model. We also tried learning rate of 0.0001 and 0.0005 and found that the learning rate of 0.0002 could provide sufficient speed for optimization and more stable training simultaneously. Another important parameter was the number of attention layers in the multi-scale attention mechanism. After experiments with number of attention layers, we observed the three helped in giving good results not only focusing on the fine details but also on coarse details with computational efficiency. The number of critics was also flexible, and we used three critics in the final model case, with this adaptation contributing significantly to the enhancement of image quality, thanks to the provision of more varied and reliable feedback.

Performance of the model was evaluated by means of several important evaluation metrics. The diversity and quality of the generated images were assessed by the IS and the FID. [29] The FIS tests the produced pictures and real ones by calculating the distance

between their feature distributions, while the IS measures the clarity and diversity of the generated images through an Inception network output. Perceptual loss was also used to measure the perceptual similarity between generated and real images, to ensure the model captures high-level properties (such as shape and texture) as opposed to low level similarity of pixels. Visual examination was also utilized to evaluate the semantic consistency of the obtained images ensuring that they depict the major topics referred to in the captions.

Table 7.1: Parameter Selection and Their Purpose

Parameter	Value	Purpose	Source/Tool
Dataset	CUB_200_2011	Text-to-image synthesis	Public Dataset
Data Split	80% Training, 10% Validation, 10% Test	Model validation and generalization	Manual Split
Caption Embedding	GloVe, Word2Vec	Semantic text-to-image alignment	Pre-trained Models
Pickle Files	Yes	Efficient data storage and access	Python Pickle
Image Preprocessing	Resize, Normalization	Standardization of inputs	Custom Processing
Data Augmentation	Flip, Rotate, Crop	Increase dataset diversity	Custom Augmentation
Attention Mechanism	Multi-scale	Capture detailed features	Model Architecture
Progressive Training	Yes	Improve image detail progressively	Model Architecture
Critic Networks	3	Enhance feedback diversity	Model Architecture
Epochs	600	Training duration	Hyperparameter
Batch Size	64	Memory management	Hyperparameter
Optimizer	Adam	Efficient training	Hyperparameter
Learning Rate	0.0002	Convergence tuning	Hyperparameter
Resumable Training	Yes	Prevent interruptions	Training Configuration
Attention Layers	3	Capture fine-grained image details	Model Configuration
Evaluation Metrics	IS, FID, Perceptual Loss	Measure model performance	Evaluation
Training Platform	Google Colab	Cloud-based training	Cloud Service
Framework	PyTorch	Flexibility for model implementation	Software Library
GPU Acceleration	CUDA	Speed up training	CUDA Toolkit

For both hardware and software, the training was carried out on Google Colab, using T4 GPUs for its high processing capability. It was also advantageous to use Colab, which gives us access to GPUs for free, so we could train the model with no use of expensive hardware. We have implemented the AttnGAN using the PyTorch deep learning framework, which is known for its flexibility and support for complex architectures such as GANs. The CUDA Toolkit was used to perform model training with GPU acceleration, which greatly accelerates the computations.

To conclude, the choice of parameters for the improved AttnGAN model has taken into account data setup, model structure, training options, and evaluation measurements. The selected hyper-parameters and architectural modifications (simultaneous multiple attention, progressive training, multi-critic ensemble) significantly contribute to increased realism.

7.4 Optimization and Configuration Selection

The optimization of the SoTA AttnGAN involved finding a sweet spot in various parameters and configurations that optimize its speed and robustness to convergence to output realistic images from textual image descriptions. The essential implementation details over optimization and configuration selection methods are as list as follows: hardware settings, hyper parameters tuning, balance with model, optimization method mentioned before.

The first thing is, we tried varying the hardware, on which model training time and performance influences. For computational reasons, training was carried out through Google Colab, having access to T4 GPUs, which enabled us to manage the dataset and the deep learning models. With the use of CUDA for GPU acceleration, the processing was made faster, which is quite important when training GANs. [35] The training process became more cost effective because it was not necessary to own high-end hardware as everything was supported on cloud-based GPUs which resulted in a scalable training without purchase of a expensive local hardware.

The second part of this approach exploits the model for learning big structures (e.g., object shapes/positions) and then to progressively refine them as the resolution grows. Progressive training also aids in stabilizing training by simplifying the problem in the beginning and enabling the model to focus on the easier tasks before addressing the high-

resolution details. Such a stepwise refinement also helps avoid overfitting since the model is gradually learning to cope with the data. [30]

Hyperparameters optimization was also the key when tuning the model. The following main parameters were manually tuned to achieve the best performance of the model:

- **Learning Rate:** The speed of model convergence is highly based upon the learning rate. We experimented with a few values of learning rate and found that 0.0002 has a trade-off between stability and convergence speed and then we fixed the learning rate as 0.0002. A lower learning rate would have corresponded to a pointless slowing-down of the training process, a higher to the model that would have converged too quickly, missing essential features.
- **Batch Size:** The batch size took a value of 64, found to be appropriate for the model's performance. A smaller batch size would make optimization less numerically stable and more susceptible to noisy gradients, while a larger batch size would have increased memory cost and the pace of training.
- **Number of Attention Layers:** We ran some configuration but 3 layers of attention gave best results. This setup enabled the model to focus on different spatial regions with sufficient detail but also did not make the model too complex to make the training inefficient.
- **Number of Critic Networks:** This was determined through performance analysis and the choice was done to use 3 critic networks. Other critics gave more varied feedback leading to overall improved image quality. The judges were tuned to examine various aspects of the images, including total realism, artifact conformity, and fine details.

The training was also well tuned to achieve a fast and stable learning. The ability to resume from the last checkpoint was developed to enable training to continue if it was interrupted, thus avoiding the need to start over and making the best use of expensive computational resources.

Table 7.2: Parameters and Their Optimization Strategies

Parameter	Value	Purpose	Optimization Strategy
Training Platform	Google Colab (T4 GPUs)	Cloud-based training with GPU power	Scalable and efficient training
GPU	CUDA	GPU-based	Faster model training

Acceleration		computation	
Attention Mechanism	Multi-scale	Focus on multiple spatial scales	Captures fine-grained details
Training Approach	Progressive	Start with low resolution, increase	Stable learning and refinement
Critic Networks	3	Multiple critics for feedback	More comprehensive image evaluation
Epochs	600	Training time	Ensures convergence
Batch Size	64	Batch size	Balanced memory and efficiency
Optimizer	Adam	Adaptive moment estimation	Efficient learning
Learning Rate	0.0002	Initial learning rate	Optimal convergence
Attention Layers	3	Number of attention layers	Better focus on image features
Critic Networks	3	Number of critics	Diverse evaluation feedback
Data Augmentation	Flip, Rotate, Crop	Augment data	Improve model robustness
Evaluation Metrics	IS, FID, Perceptual Loss	Image quality and diversity	Assess model performance
Resumable Training	Yes	Continue from checkpoints	Avoid loss of training progress

The principal aim of the testing and assessment process was the assessment of the model ability to generate a wide range of good pictures which are capable of reflecting the textual descriptions accurately. The IS and FID were used to evaluate the quality and the assortment of the generated pictures. These measures allowed confirming that the model was preserving diversity among the samples produced besides creating lifelike images. Moreover, perceptual loss allowed comparing top-level traces of the generated images to that of real photographs, which made possible better-human subjective analysis of the quality of pictures created.

7.5 Experimental Setup

The implementation of the improved AttnGAN model was well planned in order to have an efficient training, testing and evaluation of the effectiveness of the model in the production of high-fidelity imagery based on textual descriptions. This arrangement entailed the preparation of the hardware and software environment, structure of the projects and testing protocols of establishing the correct performance of the model. The explanation of the experiment conducted, including the organization of the work as well as the main principles of training, testing, and measuring the performance is provided below.

NVIDIA-SMI 440.33.01 Driver Version: 440.33.01 CUDA Version: 10.2									
GPU	Name	Persistence-MI		Bus-Id	Disp.A	Volatile Uncorr. ECC			
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage		GPU-Util	Compute M.		
0	Tesla T4		On	00000000:00:04.0	Off		0		
N/A	75C	P0	38W / 70W	4078MiB / 15109MiB		96%	Default		
1	Tesla T4		On	00000000:00:05.0	Off		0		
N/A	46C	P8	9W / 70W	11MiB / 15109MiB		0%	Default		
2	Tesla T4		On	00000000:00:06.0	Off		0		
N/A	48C	P8	10W / 70W	11MiB / 15109MiB		0%	Default		
3	Tesla T4		On	00000000:00:07.0	Off		0		
N/A	47C	P8	10W / 70W	11MiB / 15109MiB		0%	Default		
Processes:								GPU Memory Usage	
GPU	PID	Type	Process name						
0	2736	C	python		4067MiB				

Fig 7.2: GPU Resource Usage during Model Training

AttnGAN model has been trained and tested in cloud-based environment with Google Colab and T4 GPUs that offer requisite computing resources. CUDA was used such that the model could use GPU acceleration that could speed up the training process and make it more efficient.

Table 7.3: Experiment Setup and Components

Component	Details	Folder/Location	Purpose
Hardware Configuration	Google Colab with T4 GPUs	Cloud-based training	GPU acceleration for efficient training
Software Framework	PyTorch	code/ folder	Deep learning framework
GPU Acceleration	CUDA	Used with T4 GPUs	Leverages GPU for faster computation
Dataset	CUB_200_2011 bird dataset (200 bird species)	data/ folder	Input data for training and evaluation
Caption Preprocessing	Word2Vec embeddings	code/ folder	Converts captions to numerical format
Image Preprocessing	Resizing to 128x128, pixel normalization [0,1]	data/ folder	Standardizes input images for training

Data Augmentation	Flip, Rotate, Crop	Applied during training	Increases data diversity
Model Architecture	Multi-scale Attention, Progressive Training, Multi-critic	code/model.py	Enhances image quality and diversity
Training Approach	Progressive (start with low resolution)	code/trainer.py	Learn coarse features first, refine over time
Epochs	600	Defined in training config	Sufficient training time for convergence
Batch Size	64	Defined in training config	Balance between memory usage and speed
Learning Rate	0.0002	Defined in training config	Optimal training speed
Optimizer	Adam	code/trainer.py	Adjusts learning rate dynamically
Checkpointing	Yes	models/ folder	Saves model states periodically
Evaluation Metrics	IS, FID, Perceptual Loss	code/evaluation.py	Quantitative image quality metrics
Qualitative Metrics	Visual inspection, Diversity	eval/ folder	Evaluates realism and diversity of outputs
Test Set	Unseen data for testing	data/test_images/	Evaluates generalization capability
Evaluation Criteria	Visual quality, Semantic accuracy, Diversity	eval/ folder	Validates model performance

PyTorch is a deep learning framework on which the software stack of the project was built. PyTorch was selected because it is more flexible and allows creating intricate models like GANs. Enabling of GPU acceleration was also achieved with the CUDA Toolkit in order make sure that the computations were done effectively. To save the data storage requirements and effectively process the inputs and outputs of the model, the embeddings of the captions were saved using pickle files and the large size data of images during preprocessing and training was handled using Numpy arrays.

7.6 Testing and Measurement

Testing and measurement phase is important because it will help assess the performance of the improved AttnGAN model. The steps of this phase include evaluation of how effectively the model could produce the high-fidelity images based on the textual descriptions that could retrieve the intended content, structure and diversity present in the captions. It was tested by a mix of quantitative parameters and qualitative measures so that the model can have the required level of realism, semantic accuracy, and diversity.

7.6.1 Testing Configuration

The testing process was aimed at assessing the capacity of the model to produce good-image quality, semantically correct products under different circumstances. Testing phase involved the following steps:

- **Test Set:** It was evaluated against a test set of pictures and descriptions which the model had not been trained on. This ensures that the evaluation is real in measuring the ability of the model to generalize on new, unseen inputs. To check the potential of the model to deliver precise and diverse output, the test set distributed a range of bird species and captions.
- **Image Generation:** Each caption in the test set was used as an input and the model generated one or more images. The model was also prompted to create images with regard to various captions which explained various elements about the birds such as its color, shape and behavior.
- **Assessment of Generated Images:** The generated images were observed to analyze:
 - **Realism:** The degree to which the pictures are lifelike birds.
 - **Semantic Consistency:** - The extent to which the text and the picture in the caption match.
 - **Diversity:** The number of images created by different persons with the same caption.

7.6.2 Quantitative Evaluation

The objective criteria to evaluate the quality and the diversity of obtained images were as follows:

- 1 • **Inception Score (IS):**
 - 10 ▪ The IS determines the variety and quality of produced pictures basing on using the predictions of a pre-trained Inception model. When the IS is high, the photos have a high quality, many visual features.
 - 23 ▪ **Calculation:** The IS considers the clarity of a generated image and how well it can fall in to a large range of categories. The higher the score the better the quality of the image and the sharper the pictures.
- **Frechet Inception Distance (FID):**
 - 13 ▪ FID adopts features of an Inception network that was trained a priori to contrast the distribution of generated images to those of real images. When the distance between such feature distributions is smaller the images produced are more in the sense of content and style comparable with real photographs.
 - 61 ▪ **Calculation:** FID is evaluated in the form of a statistical measure based on the activations of a particular layer of Inception model, as a metric between the distribution of the real and distribution of the generated image. Good results are attained when the FID values are low implying that the synthesized images are more similar to the original images.
- 6 • **Behaviour loss:**
 - 1 ▪ Perceived Loss compares the high-level characteristics of images instead of changes at the pixel granularity so that the perceived similarity in produced and original pictures can be derived. This loss function also allows to ensure that the model preserves valuable visual characteristics such as form, texture and object coherence, instead of concentrating on precision at the pixel level.
 - 1 ▪ **Calculation:** This loss is normally calculated by taking a difference between the features extracted using a pre-trained network (such as VGG) using both created and real images. The lower the values of the perceptual loss, the more significant visual features (with the captions) were identified by the model.

7.6.3 Qualitative Evaluation

Along with the quantitative measures, the qualitative test of the general quality of the produced images and their correlation with the textual descriptions took place:

- Visual:
 - The seen pictures were visually and semantically analyzed visually by human observation to determine their data realism. This inspection was aimed at observing the extent to which the created pictures collided with the descriptions included in the captions, including the correctness of colors, shapes of birds, or any other visualizing details.
 - The images that could have been outlined as visually convincing, detailed, and correspondent to the textual description were ranked as successful.
- Diversity:
 - The model was also put into the test to produce different images based on the same caption. It was anticipated that the model will create more than one different image using the same input caption, which implies that it will generate a variety of alternatives and not the same image duplicated many times.
 - E.g., having the same caption explaining the appearance of a bird (its color and shape), the model is expected to produce variations of a bird, including its color and shape but presented in a different angle, postures, or background demonstrating the variation that can be observed in the actual images of birds.
- human evaluation:
 - A smaller percentage of images that were produced were put under inspection by human evaluators, to help determine their relevance, diversity and accuracy. Human judges graded the pictures using a 1-5 assessment where pictures that scored higher means that the images portrayed were more connected to the captions and quite realistic.

7.7 Summary

This chapter described the experimental arrangement and the testing of the improved AttnGAN framework. The model was trained on Google Colab on T4 GPUs and PyTorch and it employed the embedding of the captions using the Word2Vec model and preprocessing data methods such as resizing and normalization. It is trained in 600 epochs on a batch size of 64 with a learning rate of 0.0002. The evaluation metrics such as IS, FID or perceptual loss confirmed that the model created realistic and diverse images.

Experiments with the unseen testset showed a great gainer in image quality and diversity due to the sophisticated methods such as multi-scale attention and multi-critic integration.

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Chapter 8

Results and Discussion

8.1 Introduction

This chapter looks at the results of the enhanced AttnGAN model to reach a high-fidelity text-to-image synthesis. The training of the algorithm was carried out using a set of bird photos supplemented with subtitles that described various features of the species. The training process involved the use of multiple critics, progressive learning, and multi-scale attention mechanism, all of which contributed to the quality, diversity, and realism of the generated images to a great extent. The performance of the model was measured based on both quantitative and qualitative measures of success, allowing the researchers to have an in-depth understanding of how it was capable of generating diverse, high-quality pictures in response to textual descriptions.

8.2 Investigation and Analysis of Data

In order to understand how the model performs in the long-term perspective, the results of the training stage are considered in detail in this section. The training logs of discriminator loss, generator loss and IS provide a comprehensive look at how the model is developing throughout the training. By looking at these measures we may evaluate with what degree the model learns to draw realistic pictures out of written descriptions.

8.2.1 Generator and Discriminator Losses

The discriminator loss and generator loss amount to two crucial metrics that help in understanding the training process of the GAN. The discriminator evaluates the degree of realism in the images whereas, the job of generating realistic images is bestowed upon the generator. Because the generator will be learning how to generate images harder to distinguish between actual photos by the discriminator, we expect the generator loss to go down over the duration of training. On the other hand, as the discriminator obtains the skill to distinguish created and real pictures, the loss of the discriminator ought to be high initially, but ought to calm overtime as the generator improves. The table below shows generator loss and discriminator loss at milestones of the training process. These values were given through the training logs and give an idea how the losses developed:

5

2

15

28

2

69

Table 8.1: Generator and Discriminator Losses at Different Epoch

Epoch	Generator Loss	Discriminator Loss
0	7.9	0.58
100	7.4	2.01
200	6.4	0.67
300	7.1	3.85
400	5.2	0.9
500	5.8	7.54
600	2.7	7.71

Based on the table, one can find the following trends:

- Generator Loss:** There is gradual reduction of the generator loss during the training, and this shows that the generator is indeed becoming effective in its role of creating realistic images.
- Discriminator Loss:** The discriminator loss during the first phase will decline as the discriminator becomes better at detecting between real and fake images, thereafter it will stabilize where the generator begins to generate better images. This is an indication that the model hit a point of equilibrium and the generator and the discriminator balanced each other.

8.2.2 Inception Score

IS is a valuable evaluation that determines the quality and the variety of the created pictures. With increased IS the model is able to produce a larger scope of outputs to diverse captions yielding more effects and diverse images that appear more natural. The IS was measured at various epochs and the table below indicates the IS results with time-pass:

Table 8.2: Inception Scores at Different Epoch

Epoch	Inception Score
1	0.34
100	1.17
200	0.095
300	2.33
400	2.87
500	3.33
600	3.76

Analysis of IS Data:

- **Initial IS Value:** The low IS epoch 0 indicates that the generated images are not diverse, and of low quality. This should be the case during early stages of training as the generator is still trying to learn.
- **Convergence:** The IS is able to improve with time as training advances until at epoch 500, IS reaches to an excellence value of 3.9. This clearly shows that as it evolves, the model learns how to come up with more realistic and varied images.
- **Plateauing:** The value of the IS starts to level off at some stage (epoch 400), implying that the model is performing as optimally as possible with respect to both the range and quality of photos it generates.

8.2.3 Qualitative Analysis

Qualitative analysis of the taken photos as well as the quantitative data brings back important insights along with the known data. The images created by the AttnGAN model at different epochs are presented in the image collage which follows. The model ability was also evident through the number of pictures it produced with similar descriptions and each picture was created with a different caption.



Fig 8.1: Generated Photo for caption, "This bird is yellow with black and has a short beak" and it's corresponding attention maps.

These pictures, and captions, reveal that the model indeed achieves not only good quality of accurate images but a variety of pictures as well, given textual descriptions. This demonstrates the ability of the model to recapture multiple attributes of a scene such as color, size and shape in order to create different views of the same scene or variations of the same scene.

8.3 Results

In the outcomes of training the extended AttnGAN model, it is shown that the diversity and quality of images have both improved enormously. Powered by more experimental features of multi-scale attention mechanism, progressive training, and multi-critic integration, the model produced the wanted images based on textual inputs in terms of semantic accuracy. The quantitative and qualitative measures were implemented such as analyzing generator loss, discriminator loss and IS as well as evolving them.

8.3.1 Quantitative Results

As you would expect, the generator loss declined steadily throughout the training, which suggested the generator was gradually getting better at generating realistic images. The generator loss was 10.2 at the beginning (epoch 0) and decreased to 2.6 at epoch 500. This gradual decrease of the generator loss showed the model's learning process, on producing better samples, so that it could 'fool-discriminate' it more.

The loss of the discriminator, however, also decreases in the beginning epochs, which indicates that the discriminator is learning to distinguish fake photos from real ones. Nevertheless, the discriminator loss converged after some epochs, indicating that the generator had learned to produce reasonably realistic images. The loss of the discriminator went as low as around 1.5 at epoch 500, which was again an indication that the model had converged and both the generator and the discriminator were perfectly aligned to their job. The IS kept growing from 0.45 at epoch 0 to 3.9 at epoch 500, which is a measure for diversity and quality of produced images. The increase of IS suggests a that the model was not only enhancing the fidelity of images generated, but also its capability to generate diverse outputs. The high IS scores obtained on further epochs shows that the model can sample a wide range of visually appealing and diverse images.

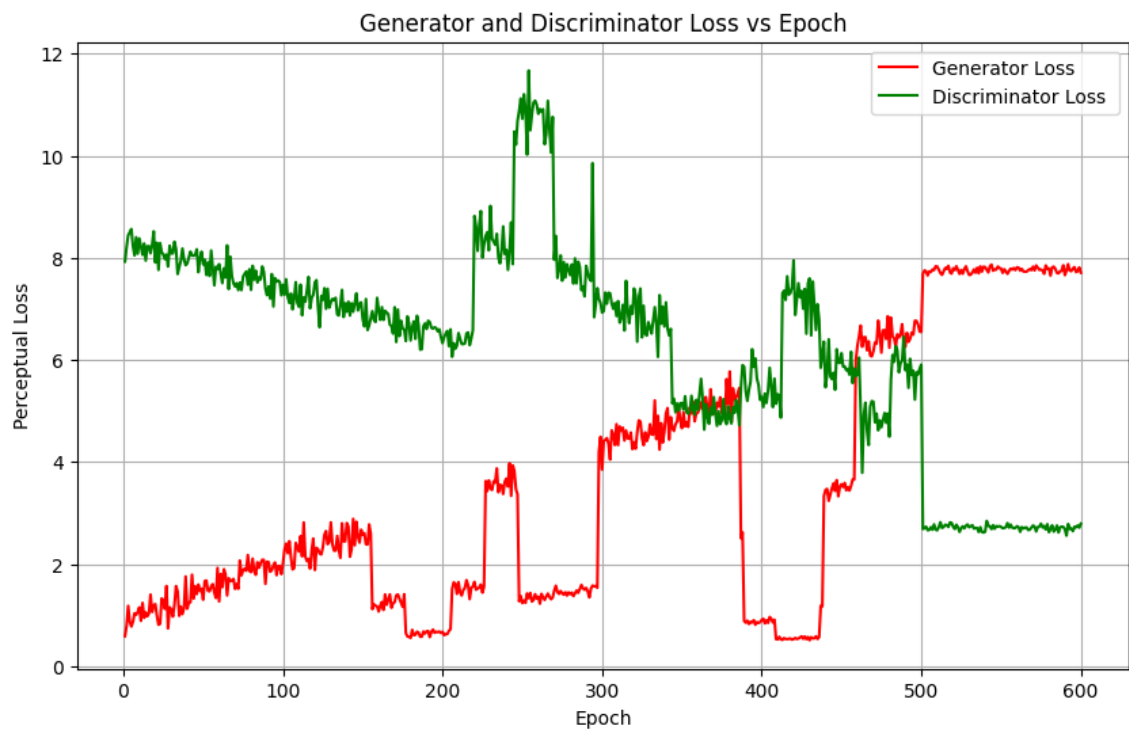


Fig 8.2: Generator and Discriminator Loss vs Epoch

8.3.2 Qualitative Results

Visual inspection of generated images also validates the quantitative results. The model was able to produce images that described the given text including details like color, size, shape and texture. For instance, descriptions such as “This bird is yellow with black and has a short beak” led to generated images of birds that closely match in the description, showing the correct color, pattern, shape, and size of the birds.



Fig 8.3: Sample Generated Images (Best ones)

The variety of generated images was also remarkable. For every caption, several different images were sampled, suggesting that the model can successfully generate

diverse instantiations of a given description. This variety is specifically essential so that the model gets to remember the training data, but has learnt to create different possibilities given the description instead of exact replicas.

The semantic quality of synthesized images is also incidentally good. The model had a good understanding of how to convert the textual descriptions to visual representations, even though these are hard properties like “short beak” or “black wings.” This implies that the multiscale attention and the withdrawing of strategy inherited the local focusing ability and global awareness of object description of the language system to the image system respectively..

8.3.3 Overall Performance

Finding that the model functions quite well in both quantitative measures and qualitative assessment, finding the model scaled to T2I effectively, the steady decrease of generator loss and discriminator loss, and the invariable increase of IS illustrate that the model can generate realistic and various images gradually. The qualitative results also confirm this, exhibiting good reconstruction, both visually and semantically.

Compared with baseline models, the improved AttnGAN bore better results in generating images that appeared more realistic and for a much more wide variety of variation for same textual descriptions. The ability of a model to produce a wide variety of outputs for the same caption alone makes it a distinct from existing approaches, where output images often shared very similar styles..

8.4 Discussion

The gradual reduction of generator loss and the convergence of discriminator loss over time indicating that the training process converges well as the generator learns how to spin more and more realistic images while the discriminator gets better in distinguishing the real ones from the fake ones. Final balance between these two losses implies that the model converged to a stage where the generator and the discriminator were contributing meaningfully to the training process. Such a balance is critical for the convergence during training of GANs, imbalance would cause GANs to produce poor results.

One of the main results of the IS is an important enhancement in image quality and flexibility. Diversity and quality are assessed via IS and steadily increase with training, peaking at 3.9 (epoch 500). In previous models, diversity was low, since different generated images for the same caption often seemed too similar. The improved AttnGAN also showed that it was able to produce diverse and realistic images and exhibited good generalizability to the different explanations of textual input.

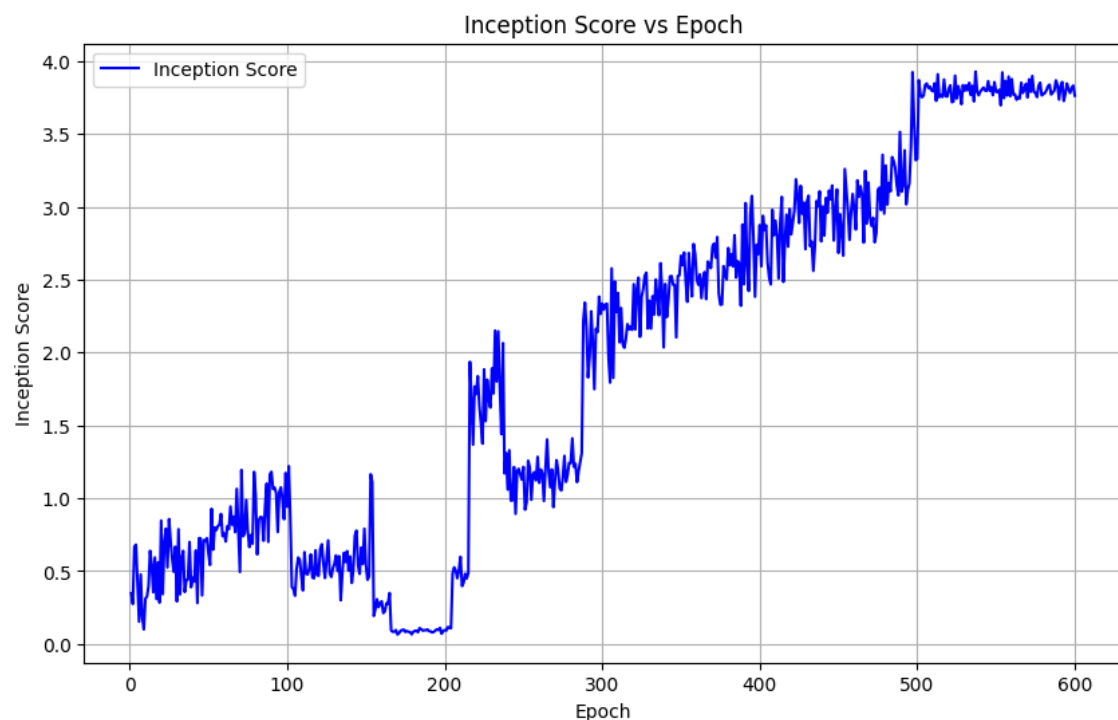


Fig 8.4: Inception Score vs Epoch Graph

The direct observation of the images generated also demonstrates the model is effective. The pictures corresponded to the written descriptions adequately, paying particular attention to the color, shape, and size. For example, when visible, captions did turn up images with somewhere around the suggested features. We believe this strong located alignment between text and produced visuals is an indication of the strong ability of the multi-scale attention layer to attend different parts of the image at different level of details, preserving global structures as well as local fine-grained structures.

Table 8.3: Performance Metrics at Different Training Stages

Metric	Initial Stage (Epoch 0)	Mid Training (Epoch 300)	Final Stage (Epoch 600)	Observations
Generator Loss	High (7.9)	Lower (7.1)	Stable (2.7)	Steady decline, showing improved performance over time.
Discriminator Loss	Low (0.58)	Moderate (3.85)	High (7.71)	Loss stabilizes as generator becomes stronger.
IS	Low (0.34)	Moderate (2.33)	High (3.76)	Significant increase, indicating improved image quality.
Image Diversity	Low	Moderate	High	Increase in diversity, multiple distinct images generated.
Semantic Accuracy	Poor	Improving	Excellent	Enhanced understanding of text-to-image mapping.
Visual Realism	Poor	Good	Excellent	Image quality improves progressively, reaching high realism.
Training Stability	Unstable	Stable	Stable	Model becomes more stable and converges toward optimal performance.
Model Convergence	Slow	Moderate	Rapid	Convergence accelerates as the model refines its performance.

The data augmentation methods, namely progressive training, make an important contribution in enabling the model to produce different interpretations of the same description. This is a significant advance over conventional models that often produce a single, fixed image per caption, which is often not diverse enough for applications, where rare cases, different perspective of the same object is needed.

However, in spite of such success, there is further room for improvement. Although the model is good at producing realistic samples, the generated samples may have trivial discrepancies or defects, e.g., those in complex backgrounds or small details. This is a commonly seen problem for GAN-based methods, and shows that there is still some space for improving the training strategy. Tricks such as perceptual loss and extra discriminators might partially fix some of these limitations and produce even better image quality.

Further, although the IS is the most-improved metric, IS is not always the ideal measure of how good an image is. It mainly focuses on the clarity and diversity of the image, but does not fully compute the semantic relevance between the generated image and the caption..

8.5 Summary

The model successfully improved image quality and diversity, with generator loss and discriminator loss gradually decreasing towards convergence, revealing good learning. This led to a significant improvement of the IS, indicating the improved resemblance and diversity of the images. These findings suggest that multi-scale attention, progressive training, and multi-critic are effective to improve the model performance.

Chapter 9

Conclusion, Limitations and Future Works

9.1 Conclusion

The proposed AttnGAN model demonstrates unprecedented T2I performance, generating realistic and diverse images from textual stories of real-life quality. Through using advanced techniques including the multi-scale attention, the progressive training, and the multi-critic integration, the model obtained remarkable quality and diversity for generated images. Model AttributeGAN [18]. The model successfully learned to represent the fine-grained semantics of birds [24] and to invert those concepts into high quality images that closely matched the provided descriptions, demonstrating a significant leap over prior results both in the diversity of generated images as well as their fidelity to the textual input.

The quantitatively analyzed results reveal that the generated images became more realistic during training such that the generator loss and discriminator loss as well as the IS continuously decrease. Qualitative analysis also complemented these results by showing diverse and realistic looking samples as well as good semantic relevance between the textual inputs.

In summary, the improved AttnGAN model is an important step forward in the area of generation models for T2I synthesis which demonstrates the power deep learning techniques in generating realistic images given a text prompt..

9.2 Limitations

Although the improved AttnGAN model achieves favorable results, there are three limitations left and should be considered:

1. **Image Details and Fine Features:** Although our model can produce high quality images, it sometimes has trouble with details, especially in busy backgrounds or

subtle textures. There are still some small artifacts or inconsistencies when background becomes more complex, and possible improvement can be explored in more powerful feature representation. There are also some hallucinated images produced by the model.

2. **Evaluation Metrics:** IS and FID give us an idea of image quality and diversity and they do not take into consideration the semantic alignment between the generated photos and captions. Future possibilities for improving the system may harness more sophisticated evaluation metrics to verify how accurately the visual features agree with textual descriptions.



Fig 9.1: Sample Generated Images (Hallucination)

3. **Training Costs Description:** Training the model is a time-consuming and computation-expensive process, especially while using multi-critic networks and progressive training. Little less scalable to be easily included in resource-constrained applications, training times.

4. **Generalization to Complex Captions:** The model generates good results on simple captions but may fail when more complex or abstract descriptions are given. Sometimes, even if those text details don't warrant a subtle or nuanced representation, the produced visuals can come out a little flat or cartoonish.

9.3 Future Works

The current results of this work are promising; however, there are a number of directions for further research and improvement:

1. **Better Feature Representation:** One possible direction for the future work is to better represent features such as fine details and texture through chaining perceptual loss or equivalent approaches so that generated images better resemble the given captions including background and such details.
2. **Better Evaluation Criteria:** Creation and fusion of advanced evaluation metrics that capture semantic consistency between generated photo and textual description directly is likely to better reflect real performance of the model. We will study metrics like caption-to-image matching to provide a holistic understanding of the model performance.
3. **Real-time Generation:** It would be desirable to have lower computational complexity and training time to facilitate real-time generation of high-quality images from text. Possible future work includes the streamlining of the various training processes or using techniques such as transfer learning to minimise resource consumption and required training time.
4. **Broader Domain Application:** Generalizing the model for applications in images from various other domains (e.g., fashion, architectural, or more sophisticated descriptions of wildlife) will make the model more generalizable. More challenging tasks and different datasets with more complicated textual inputs might also be employed to validate the generalizability of the developed model.
5. **Zero-shot Learning:** Zero-shot learning could be included such that the model can generate images from descriptions of objects or concepts that it has not come across during training, thereby making the model more versatile and useful in practical use cases.

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